

Export Controls, Shared Suppliers, and Competitive Reallocation in AI Chips

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O1 Event-Study Robustness: Alternative Designs and Sample Restrictions

Data Appendix

The descriptive exhibits and event-study designs in Section [O1](#) use monthly Korean customs export data from the Korea Trade Statistics Promotion Institute (TRASS), organized by HS10 product code and partner country, from January 2010 to the latest available month. For each destination-item pair $i = (c, g)$ and month t , the data report export values (USD) and export quantities (kg).

Variable construction. For each cell, I compute unit values $UV_{it} = \text{USD}_{it}/\text{kg}_{it}$, dropping cells with zero quantity. I construct 12-month trailing moving averages for all value, quantity, and unit-value series. Item shares (e.g., MCP–IC’s share of Korea’s total exports) use the world total row (CountryCode = WR) as the denominator. Bilateral decompositions compare Taiwan to the rest of the world, where “Others” equals WR minus Taiwan.

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Index normalization. For figures that show indexed unit values, each item’s UV is normalized to its January 2019–July 2022 mean (pre-policy = 100), so that levels are comparable across items with very different USD/kg magnitudes (e.g., MCP–IC peaks near 150,000 USD/kg while commodity DRAM/Flash are an order of magnitude lower).

Policy and technology overlays. Vertical markers and shaded windows indicate: (i) HBM3/H100 production ramp (October 2021), (ii) the October 2022 U.S. export-control regime (Entity List expansion and FDP Rule), and (iii) the October 2023 comprehensive update. These dates are taken from publicly available regulatory announcements and are aligned with the chronology in Section O1 and Table O1.2.

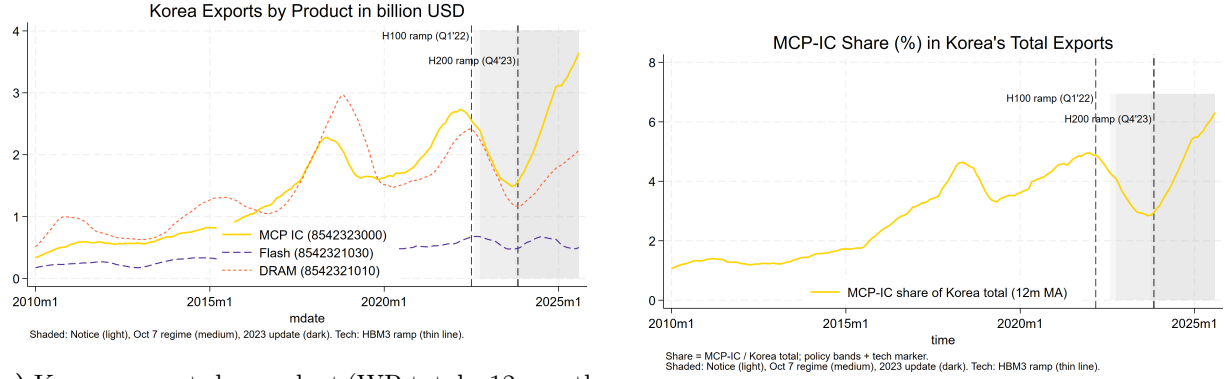
Additional Descriptive Evidence

Figure O1.1 establishes the aggregate scale of the MCP–IC boom by placing MCP–IC alongside commodity memory in Korea’s world-total exports and reporting its share of Korea’s total exports. The series move together through 2023 and then diverge as MCP–IC rises sharply from 2024 while DRAM and Flash recover more modestly; by 2025, MCP–IC approaches 6% of Korea’s total exports. This scale evidence is descriptive and does not separate technology-cycle and policy timing.

Figure O1.2 compares Taiwan-bound customs unit values across MCP–IC, DRAM, and Flash. The MCP–IC series diverges sharply during the AI boom, but the aggregate customs ratios do not recover bilateral negotiated prices; Remark 1 states the relevant interpretation limits.

Remark 1 distinguishes the customs unit values in Figure O1.2 from the model’s unobserved bilateral link prices.

Remark 1 (Aggregate unit values vs. bilateral link prices: three caveats). *The customs data record destination-level unit values, not the bilateral negotiated prices p_{du}^U of the model. Aggregation across buyers, HBM generation and quality composition, packaging intensity,*



(a) Korean exports by product (WR totals, 12-month moving average). Shading marks the policy windows; the thin line marks the HBM3/H100 ramp.

(b) MCP-IC share of Korea's total exports (WR HS=0, 12-month moving average).

Figure O1.1: Aggregate scale of the MCP-IC boom in Korean exports. The panels compare world-total exports across products and report MCP-IC's share of Korea's total exports. Both panels are descriptive.

and legal constraints on resale can all create destination-level dispersion. The bilateral link prices are unobserved, so these descriptive series cannot recover them or establish the terms of particular contracts.

Event-Study Profiles Around the Policy Dates

This subsection traces the single focal Taiwan-MCP-IC corridor around the initial controls in October 2022 and the comprehensive update in October 2023. Figure O1.4 reports event-time profiles, and Table O1.1 reports pooled contrasts using the fixed effects and residualization summarized in the figure caption. Rejected full-window pre-trends and one focal unit make the exercise descriptive rather than causal. The structural model therefore treats link-specific price responses as comparative statics at calibrated primitives, not quantities recovered from these data.¹

I use the Taiwan-MCP-IC export flow as the focal unit. Although the statutory controls apply to sales to Chinese destinations, Taiwan is informative because it is a major site where HBM is integrated into AI accelerators; the corridor is not a direct measure of statutory

¹For an overview of U.S.-China technology competition and export control policy, see [Branstetter \(2024\)](#). A related policy setting is Japan's 2019 export restrictions on semiconductor materials to Korea, studied by [Makioka and Zhang \(2024\)](#).

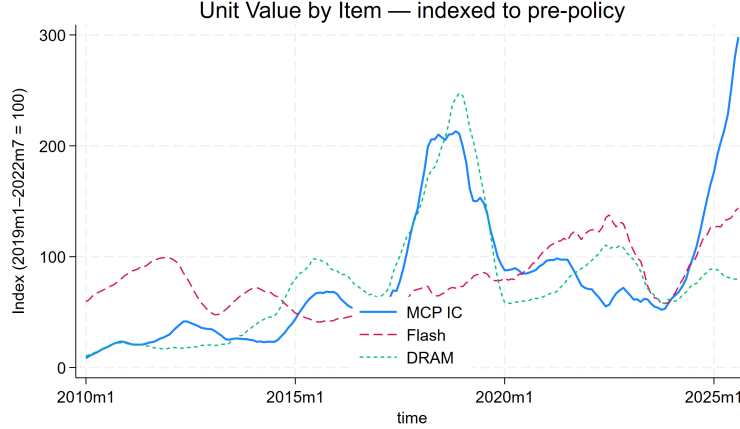


Figure O1.2: Customs unit values of Korean exports to Taiwan by item (MCP-IC, DRAM, Flash; USD/kg, 12-month moving average). These destination-level ratios are descriptive and are not bilateral negotiated prices.

exposure. Figure O1.3 documents the destination divergence: Korea’s MCP-IC exports to China and Hong Kong together accounted for 29–37% of the total in 2020–2022, while Taiwan’s share was only 7–11%. After October 2022, Taiwan’s share rose to 36% by 2024 and 52% by 2025, while the China-plus-Hong-Kong share fell to 14% and then 9%. This compositional shift spans both the policy episode and the technology transition.

Commodity DRAM and NAND serve as comparison series because they share upstream wafer inputs but not the advanced-packaging constraints of the AI vertical. Rejected pre-trends rule out a credible no-policy counterfactual. Two alternatives that avoid this cross-product benchmark—within-MCP-IC across destinations, and within Taiwan across non-AI HS codes—produce profiles similar in sign and magnitude in Section O1, specifically Figures O1.6 and O1.7.

I account for observed technology-cycle covariation through residualization. The specification includes destination-by-month and item-by-month fixed effects and an AI-demand proxy (NVIDIA data-center revenue) interacted with pre-period exposure. Because the proxy may itself respond to the controls, residualization can absorb part of the movement of interest; the HBM3E generational step and the CoWoS capacity ramp are only partially absorbed. See the [stacked](#), [restricted-sample](#), and [narrow-window](#) variants below.

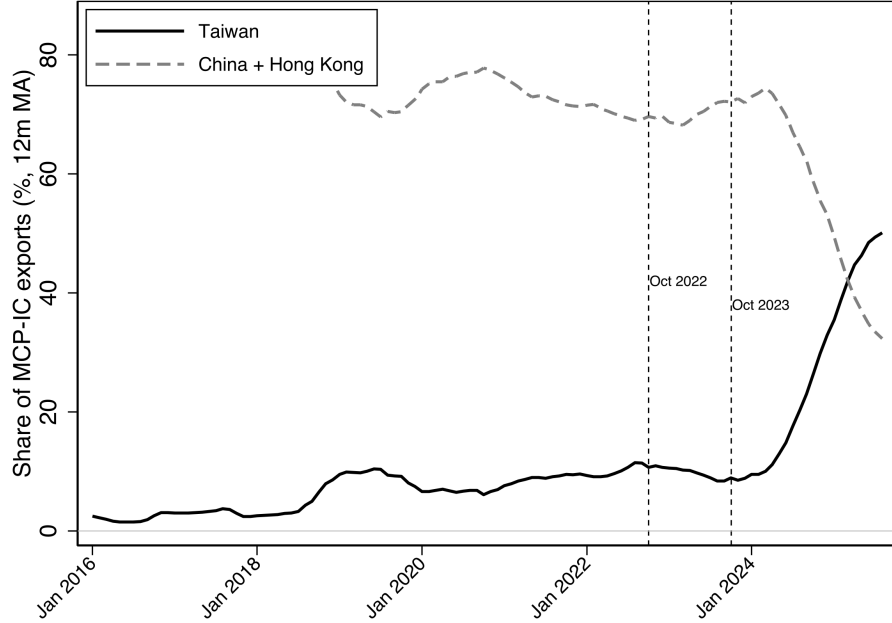
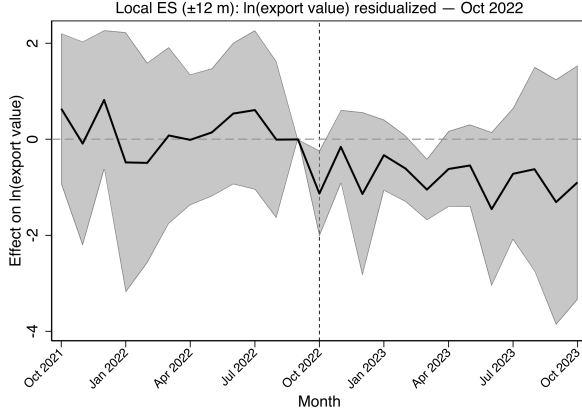
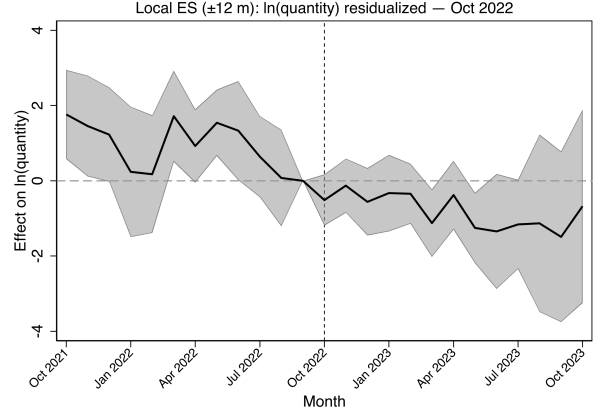


Figure O1.3: Korea’s MCP–IC exports by destination: Taiwan vs. China + Hong Kong (share of total, 12-month moving average). Vertical dashed lines mark the October 2022 and October 2023 U.S. export-control events. The series describe destination composition around the two dates; they do not identify a policy effect.

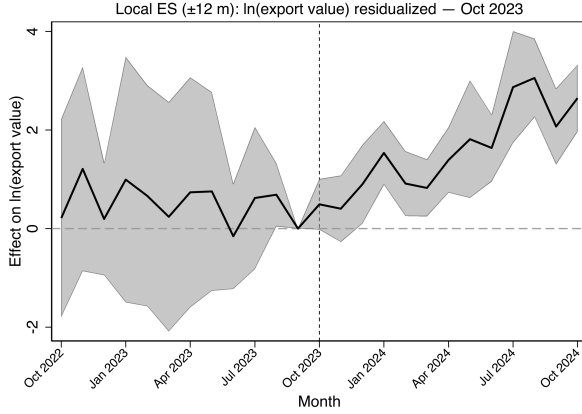
Figure O1.4 reports the profiles. The pooled Focal×Post contrast is about 55% in value and 80% in quantity after October 2022 and changes sign after October 2023 (Table O1.1). Pre-trends are rejected ($F(11, \cdot) = 4.63$ for value and 18.86 for quantity, $p < 0.001$), coinciding with the HBM3 ramp. The 2022 quantity post-minus-pre statistic is unusually extreme across assignments ($p = 0.019$; Section O2). Narrowing the pre-window flattens the value pre-trend ($F(6, \cdot) = 1.31$, $p = 0.268$) but leaves the quantity pre-trend, consistent with front-loading (Section O1).



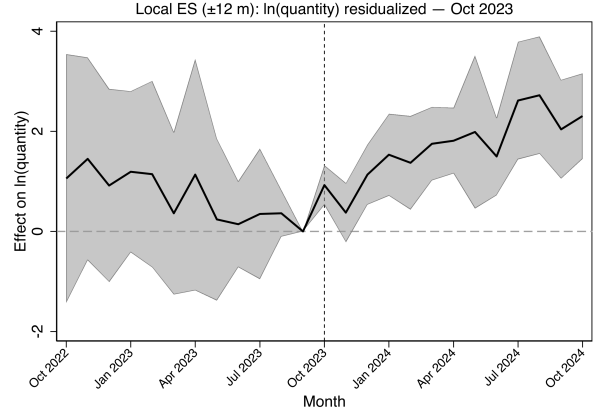
(a) Local event study (Oct 2022): $\ln(\text{export value})$.



(b) Local event study (Oct 2022): $\ln(\text{quantity})$.



(c) Local event study (Oct 2023): $\ln(\text{export value})$.



(d) Local event study (Oct 2023): $\ln(\text{quantity})$.

Figure O1.4: Local ± 12 month event studies for the single Taiwan $\times$ MCP-IC focal cell around the two policy dates. Outcomes are residualized for unit fixed effects, year-month fixed effects, unit trends, and unit-specific seasonality; regressions add destination-by-month and item-by-month fixed effects and use inverse-value weights. Unit-clustered confidence bands reflect one focal unit, and pre-period movements coincide with technology-cycle dynamics in Section O1. The [stacked](#) and [restricted-sample](#) variants report the corresponding designs.

The October 2023 export-value and quantity panels in Figure O1.4 change sign: Taiwan–MCP–IC quantities and export values rise relative to the comparison series. Read with Figure O1.3, this co-movement documents reallocation toward Taiwan. Table O1.1 reports pooled post-event contrasts for both regimes and outcomes. Section O2 separately defines the Fisher permutation statistic and reports its results; those p -values do not apply to the Focal×Post coefficients in Table O1.1.

Together, Figures O1.3 and O1.4 motivate the concentrated vertical-network setting in Section 2: flows and aggregate unit values differ across destinations around the two policy dates. They neither reveal bilateral link prices p_{du}^U nor enter the calibration; the counterfactual responses in Section 5 are conditional model predictions.

Event Study: Regression Estimates

Table O1.1 summarizes the event-study estimates in a compact pooled-contrast format. Each column reports the Focal × Post coefficient, where Focal is the Taiwan × MCP–IC indicator and Post equals one for the twelve months following each policy event ($k \geq 1$). All specifications absorb unit, destination-by-month, and item-by-month fixed effects and include the AI-demand control ($AI_t \times W_i$). Because there is a single focal unit, the unit-clustered standard errors are not asymptotically valid for inference, and the table carries no significance stars. Section O2 and Table O2.1 instead report Fisher inference for a distinct post-minus-pre statistic; because it compares the post-event mean with the full pre-window, it should not be conflated with the Focal×Post coefficient.

Columns 1–2 of Table O1.1 show negative pooled contrasts after October 2022, while columns 3–4 show positive contrasts after October 2023. The opposite signs coincide with the policy and industry changes in Section 4 and the technology chronology in Table O1.2. Figure O1.4 shows the full dynamic profile for each event and outcome; Table O2.1 reports the separate permutation results.

Table O1.1: Pooled post-event contrasts for the single focal Taiwan $\times$ MCP–IC export corridor.

	Oct 2022 Controls		Oct 2023 Controls	
	(1)	(2)	(3)	(4)
	ln(value)	ln(qty)	ln(value)	ln(qty)
Focal $\times$ Post	-0.808 (0.282)	-1.621 (0.206)	1.157 (0.530)	1.045 (0.367)
Observations	1,250	1,250	1,237	1,237
Within R^2	0.007	0.028	0.023	0.017

Notes: Unit, destination-by-month, and item-by-month FEs absorbed.

Standard errors in parentheses, clustered by unit (not asymptotically valid with a single focal unit).

Clustered standard errors are not used for inference.

Fisher p -values in Table O2.1 test a distinct post–pre statistic, not these coefficients.

Focal: Taiwan $\times$ MCP–IC. Post: months $k \geq 1$ within ± 12 window.

All outcomes residualized for unit trend and unit-by-month seasonality.

AI control: demeaned log NVIDIA data-center revenue $\times$ unit exposure weight.

Technology upgrade timeline

Table O1.2 lists major memory and packaging upgrades between 2021 and 2025—DDR5 and LPDDR5T on the DRAM side, successive HBM3/HBM3E ramps, and steep layer-count progressions in 3D NAND. I include vendor-stated or directly implied percentage improvements in speed, bandwidth, efficiency, or density to give a sense of magnitudes. The purpose of the table is twofold. First, it documents technology shocks that could move prices or quantities independently of policy, clarifying what “parallel trends” should (and should not) look like for the comparison series. Second, it provides precise markers overlaid in the event-study figures to visually distinguish policy windows (Aug–Oct 2022 notice; Oct 7 2022–Oct 2023 regime; Oct 2023 update) from contemporaneous technology milestones (e.g., HBM3E commercialization). Because MCP–IC is the customs line that records HBM-attached packages, its behavior is expected to be more tightly coupled to HBM3/HBM3E timing than DRAM (commodity DDR5/LPDDR) or NAND; the table makes those relative exposures transparent.

This context helps interpret the event-study results in Section 4. The design contrasts

Table O1.2: Major memory and packaging technology upgrades, 2021–2025

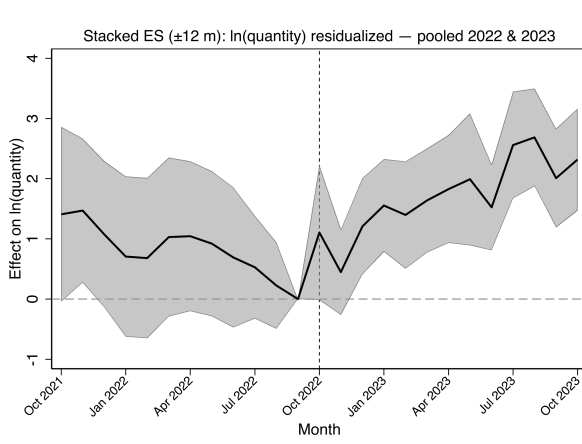
Date	Segment	Generation / Node	Key spec	Noted improvement
Dec 2022	DRAM (DDR5)	Samsung 12nm-class 16Gb	Up to 7.2 Gb/s	23% better power efficiency vs prior gen (vendor) ^a .
Jan 2023	DRAM (LPDDR)	SK hynix LPDDR5T	9.6 Gb/s/pin	+13% speed vs LPDDR5X 8.5 Gb/s (vendor).
Oct 2022	HBM	HBM3 (mass prod. reported)	~6.4 Gb/s/pin; up to ~819 GB/s per stack	HBM3 vs HBM2E: higher pin rate/bandwidth (context).
Mar 2024	HBM	Samsung HBM3E (mass prod.)	9.8 Gb/s/pin; up to 1.25 TB/s per stack	+53% per-pin rate vs HBM3 6.4 Gb/s (calc.).
Mar 2024	HBM	Micron HBM3E (volume)	up to ≥ 1.2 TB/s per stack	Power efficiency and bandwidth uplift vs HBM3 (vendor).
Jul 2022	NAND	Micron 232-layer (V8)	232L 3D NAND	Industry-first 232L; perf./power gains vs prior gen (vendor).
Aug 2022	NAND	SK hynix 238-layer	238L 3D NAND	“World’s highest” layers at the time (vendor).
May 2024	NAND	Samsung V9 (236-layer)	236L 3D NAND	Read speed up to 2.4 Gb/s; >10% energy efficiency vs V8 (vendor).
Jan 2025	NAND	SK hynix 321-layer (ann.)	321L 3D NAND	Next-gen density milestone (vendor).

Notes: “calc.” indicates a simple calculation using vendor-stated numbers (e.g., $9.8/6.4 - 1 \approx 53\%$).

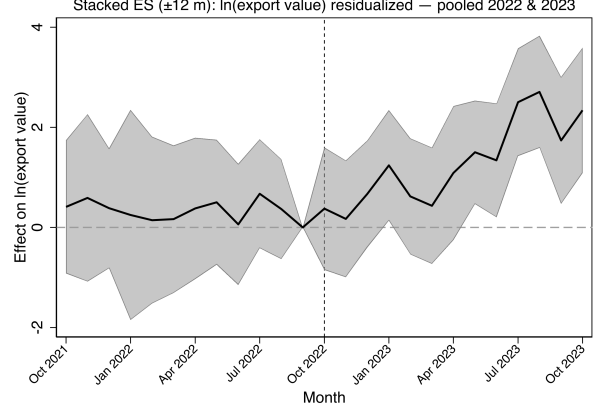
Specs and improvements are as reported by vendors unless otherwise stated.

^a Samsung’s 12nm-class DDR5 PR highlights both the up-to-7.2 Gb/s data rate and 23% power-efficiency improvement.

Taiwan×MCP–IC (the focal unit) against two sets of comparisons: (i) the same item shipped to other destinations, and (ii) other memory items (DRAM, NAND) shipped to Taiwan. The timeline shows why DRAM and NAND are informative comparisons: they experienced meaningful—but differently timed—technology progress (e.g., DDR5 efficiency, NAND layer counts) that should influence prices and volumes, yet they lack the same dependence on advanced-packaging bottlenecks and HBM qualification that bind MCP–IC to the accelerator chain. Taiwan-directed MCP–IC premia and volume shifts concentrated around the policy windows are consistent with reallocation toward the TSMC+CoWoS+HBM hub, but the same period also contains major technology and capacity changes. Placing technology markers on the x-axis, residualizing outcomes for seasonality, unit trends, and an interacted AI-market proxy, and reporting local windows separately from stacked windows make those competing movements more visible; they do not isolate a policy effect. Remaining caveats include the vendor-reported nature of some specs, potential lags between disclosure and effective line ramp, and the coarse mapping from HS codes to narrowly defined HBM products; the [stacked](#), [restricted-sample](#), and [narrow-window](#) variants address these concerns.



(a) Stacked event study for residualized $\ln(\text{quantity})$, pooled across the 2022 & 2023 event windows.



(b) Stacked event study for residualized $\ln(\text{export value})$, pooled across the 2022 & 2023 event windows.

Figure O1.5: Stacked local event studies (± 12 months) around Oct 2022 and Oct 2023 for the single focal Taiwan $\times$ MCP-IC cell. For each event, I create a separate panel and estimate with unit-by-event fixed effects and destination-by-month-by-event and item-by-month-by-event fixed effects. Outcomes are residualized for unit trends, unit-specific seasonality, and an interacted AI-market proxy. Shaded bands denote 95% CIs. The pooled profiles are descriptive and do not identify causal effects.

Stacked event studies

Figure O1.5 complements the single-event specifications by pooling the Oct 2022 and Oct 2023 policy dates in a stacked event study. For each event, I create a separate relative-time window of ± 12 months and treat the same destination-item cell appearing in two windows as two distinct panels. Estimation includes unit-by-event, destination-by-month-by-event, and item-by-month-by-event fixed effects, with the outcomes already residualized for unit trends, unit-specific seasonality, and the interacted AI-market proxy. The stacked profile summarizes the short-run comparisons around both dates in one display.

A concern with using a single long window is that medium-run technology cycles and unrelated shocks can distort pre- and post-date contrasts. I limit this problem by (i) centering on each regime change separately; (ii) re-indexing panels by event so unit composition and fixed effects can differ across windows; and (iii) retaining destination $\times$ month and item $\times$ month fixed effects within event. These choices absorb observed seasonality and

commodity-wide movements, but they do not restore parallel trends or isolate policy variation. The stacked specification therefore asks only whether the focal Taiwan×MCP–IC cell moves differently from the comparison cells, on average across the two windows.

The stacked plots in Figure O1.5 show muted visual pre-trends in the 12-month leads, although the full-window joint F -test rejects the null as reported in Section O1, followed by a rise in both quantity and export value after the event dates. The quantity and value profiles remain elevated over the subsequent year. This co-movement is consistent with reallocation toward the Taiwan hub, but the stacked design does not establish packaging scarcity, bargaining responses, or a policy cause.

Stacking averages heterogeneous movements across the two regimes, so pooled coefficients blend the larger October 2023 shift with the earlier profile. Confidence bands remain wide at longer horizons because each window contains few months. Residual differential shocks correlated with the focal-cell indicator can still affect the contrasts despite the AI-proxy residualization and fixed effects. I therefore retain Figure O1.5 as a descriptive robustness summary; Section O1 reports the two local windows separately.

Event studies with restricted samples

To complement the [local](#) and [stacked](#) full-sample designs, I estimate two restricted-sample event studies that make the comparison groups more transparent. In both designs, I use residualized outcomes. I remove unit fixed effects and HS-specific month-of-year seasonality separately from log export value and log quantity, then use the residuals as the dependent variables in the event-time regressions.

Design A keeps only the MCP–IC item and compares Taiwan to other destinations. The unit of observation is a destination-month; Taiwan is the focal destination, and other destinations that import MCP–IC from Korea provide the comparison series. Design B keeps only Taiwan and compares MCP–IC to DRAM/NAND within Taiwan. Here, the unit of observation is an HS-code-month; MCP–IC is the focal series, and DRAM and NAND

exports to Taiwan provide the comparisons.

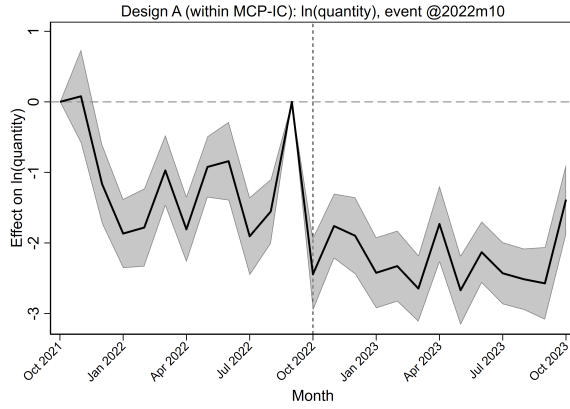
For each design and each policy date $T_0 \in \{2022m10, 2023m10\}$, I restrict the sample to a symmetric ± 12 -month window around T_0 and estimate a saturated TWFE event study,

$$y_{it} = \sum_{k=-12, k \neq -1}^{12} \beta_k \mathbf{1}\{t - T_0 = k\} \times \text{Focal}_i + \alpha_i + \mu_t + \delta_1 \text{AIxW}_{it} + \delta_2 t + \delta_3 t \times \text{Focal}_i + \nu_{it}, \quad (\text{O1.1})$$

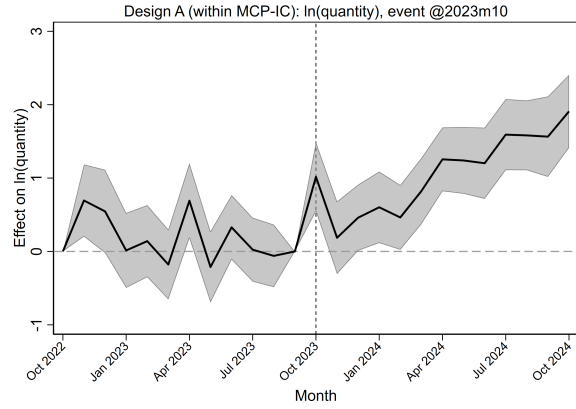
where i indexes units (destination-item cells in Design A and HS-item cells in Design B), α_i are unit fixed effects, μ_t are month fixed effects, and AIxW_{it} is the AI-exposure control described in Section 4. I omit $k = -1$ as the reference month, so all β_k are interpretable as deviations from the month immediately before the policy change. Robust standard errors are used throughout. In Design A, I also reweight comparison destinations using overlap weights constructed from pre-period means and linear trends of the residualized outcomes; these weights improve balance in the pre-window while keeping the average weight equal to one. Design B keeps uniform weights because the number of items within Taiwan is small and already quite balanced.

Figure O1.6 reports Design A. After the October 2022 date, Taiwan's MCP-IC exports fall by roughly 85–90% relative to other destinations, with smaller pre-period movements. Around the October 2023 date, the profile reverses and trends upward over the following year, reaching a threefold-to-sevenfold rise in quantities and more than a tenfold rise in values by late 2024. The pattern is consistent with reallocation toward Taiwan, but the comparison does not assign that movement to the controls.

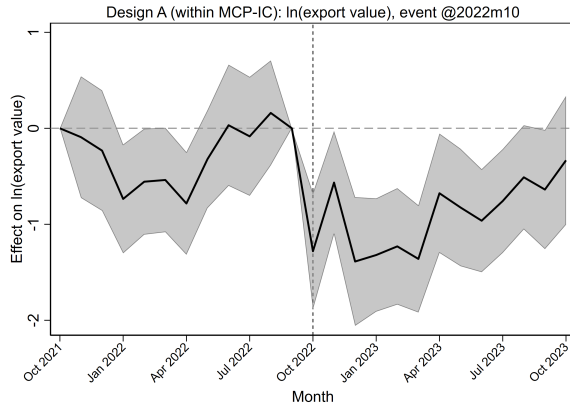
Figure O1.7 reports the corresponding within-Taiwan comparison. After the 2022 date, MCP-IC exports fall relative to DRAM/NAND, with modest visual pre-trends; after the 2023 date, they rise in both quantity and value.



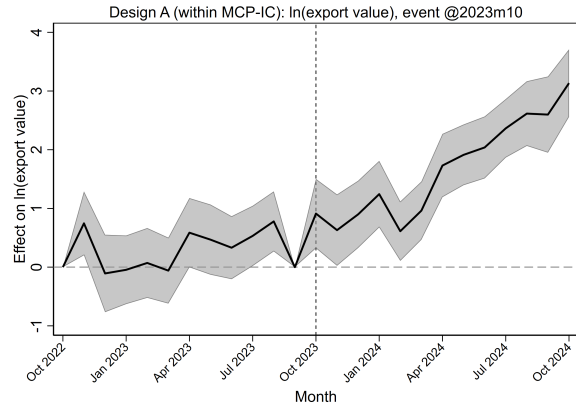
(a) $\ln(\text{quantity})$, event at 2022m10



(b) $\ln(\text{quantity})$, event at 2023m10

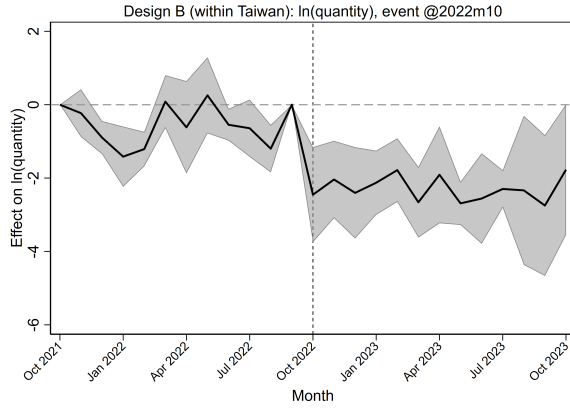


(c) $\ln(\text{export value})$, event at 2022m10

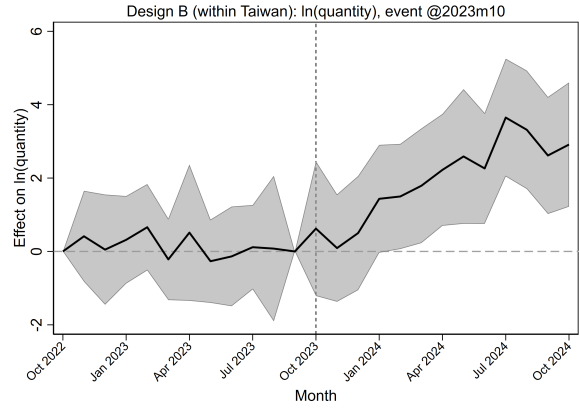


(d) $\ln(\text{export value})$, event at 2023m10

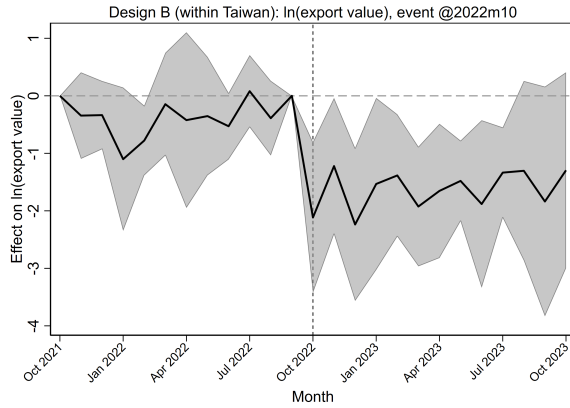
Figure O1.6: Event studies with restricted samples, Design A (within MCP-IC across destinations). The focal series is MCP-IC exports to Taiwan; comparison series are MCP-IC exports to other destinations. Outcomes are residualized as defined in Section O1. Shaded bands show 95% confidence intervals. The profiles are descriptive and do not supply a causal counterfactual.



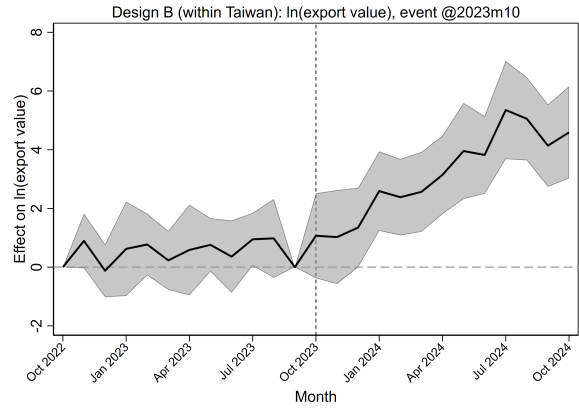
(a) $\ln(\text{quantity})$, event at 2022m10



(b) $\ln(\text{quantity})$, event at 2023m10



(c) $\ln(\text{export value})$, event at 2022m10



(d) $\ln(\text{export value})$, event at 2023m10

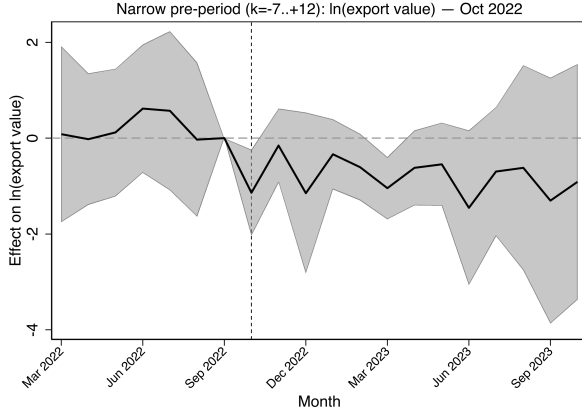
Figure O1.7: Event studies with restricted samples, Design B (within Taiwan across items). The focal series is MCP-IC exports to Taiwan; comparison series are DRAM and NAND exports to Taiwan. Outcomes are residualized as defined in Section O1. Shaded bands show 95% confidence intervals. The profiles are descriptive and do not supply a causal counterfactual.

Narrow pre-period robustness

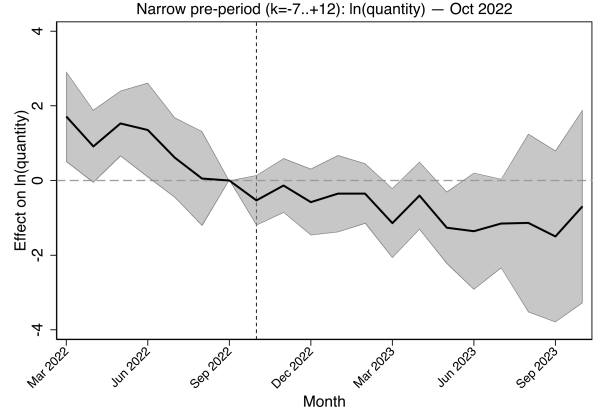
The full-window pre-period F-tests reported in Section O1 reject parallel trends for both export value and quantity around both policy events. The early pre-period months ($k \leq -8$) coincide with the HBM3 development ramp and with a rise in MCP-IC relative to commodity memory before the policy dates (Section O1). To assess how the pooled post-event summary changes when these early leads are omitted, Table O1.3 reports the mean post-event coefficient under a full window ($k = -12$ to -2) and a narrow window ($k = -7$ to -2); in both cases, coefficients are measured relative to the omitted month $k = -1$. Rejected pre-trends preclude interpreting that statistic as a causal effect.

The mean post-event coefficients are similar across the two specifications for all four outcome–event combinations: the largest absolute difference is 0.056 in logs (Oct 2023 quantity), a few percentage points and smaller than the standard error for either specification. The narrow pre-period F-statistic for export value in October 2022 is $F(6, \cdot) = 1.31$ ($p = 0.268$), so the shorter-window value pre-profile is flatter after the HBM3 ramp months are excluded. This comparison does not repair the rejected full-window pre-trends or convert the profiles into causal estimates.

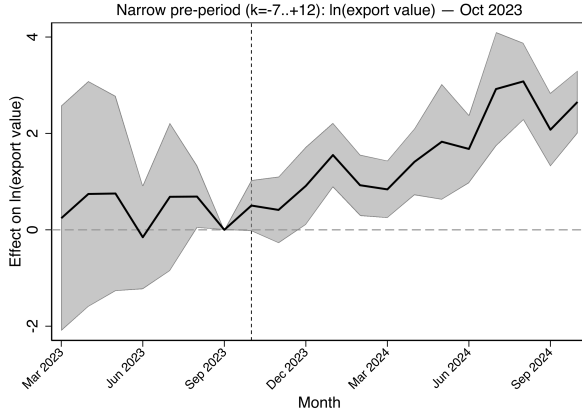
Figure O1.8 plots the narrow-window ($k = -7$ to $+12$) event studies alongside the full-window ($k = -12$ to $+12$) baseline. The post-event profiles are similar across specifications; the narrow-window pre-period confidence bands are tighter and closer to zero.



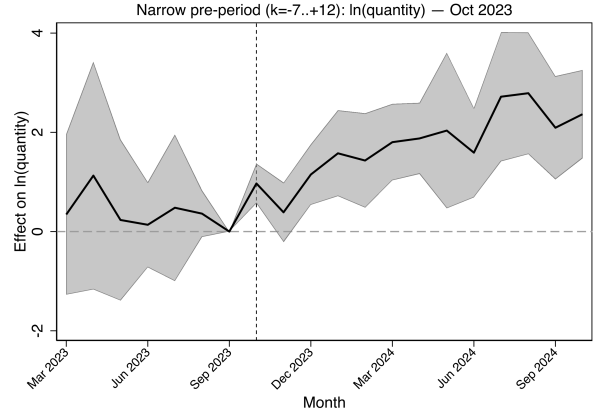
(a) Narrow window, Oct 2022: $\ln(\text{export value})$.



(b) Narrow window, Oct 2022: $\ln(\text{quantity})$.



(c) Narrow window, Oct 2023: $\ln(\text{export value})$.



(d) Narrow window, Oct 2023: $\ln(\text{quantity})$.

Figure O1.8: Event studies with narrow pre-period ($k = -7$ to $+12$). The specification is identical to Figure O1.4 except the window extends only 7 months before each event, excluding the HBM3 development ramp months ($k \leq -8$). Post-event profiles are similar to the full-window baseline, but the rejected full-window pre-trends continue to preclude a causal interpretation.

Table O1.3: Mean post-event coefficients: full vs. narrow windows

Pre-period	Oct 2022		Oct 2023	
	ln(value)	ln(qty)	ln(value)	ln(qty)
Full ($k=-12$ to -2)	-0.814 (0.602)	-0.804 (0.568)	1.582 (0.323)	1.696 (0.390)
Narrow ($k=-7$ to -2)	-0.813 (0.604)	-0.814 (0.579)	1.597 (0.327)	1.752 (0.417)
Pre-trend F (full)	4.63 [< 0.001]	18.86 [< 0.001]	6.62 [< 0.001]	5.60 [< 0.001]
Pre-trend F (narrow)	1.31 [0.268]	6.46 [< 0.001]	4.42 [0.001]	2.30 [0.048]

Notes: Entries average post-event coefficients ($k = 0, \dots, 12$) relative to omitted $k = -1$.

Full pre-period: $k = -12$ to -2 (11 coef.); narrow: $k = -7$ to -2 (6 coef.).

Standard errors in parentheses; p -values in brackets. Specifications include

destination $\times$ month, item $\times$ month FE, unit trend, seasonality, AI demand control; inverse-value weighted.

O2 Permutation / Placebo Inference

With a single focal unit (Taiwan $\times$ MCP-IC), standard asymptotic inference based on clustered standard errors is unreliable. I implement a Fisher permutation test following [Bertrand et al. \(2004\)](#) and [Conley and Taber \(2011\)](#). Eligibility requires at least 24 non-missing monthly observations for each destination and product cell. The reference set contains 54 cells for the 2022 event and 53 for the 2023 event. For each comparison unit (country $\times$ HS-code cell), I assign focal status to that unit and compute the same post-minus-pre residualized mean, denoted $\hat{C}$. Table [O2.1](#) reports this descriptive contrast directly; the axes in Figure [O2.1](#) denote it by ATT, but it is not interpreted as a causally identified treatment effect. The empirical p -value is the fraction of comparison-unit statistics with $|\hat{C}_{\text{comparison}}| \geq |\hat{C}_{\text{focal}}|$.

Table [O2.1](#) reports the results. The October 2022 quantity post-minus-pre statistic ($\hat{C} = -1.09$ in logs, a 66% decline) is unusually extreme among comparison-unit assignments ($p = 0.019$): 1 of 54 assignments has a contrast at least as large in absolute magnitude. This

permutation statistic differences the post-event mean against the full pre-event window and is smaller in absolute magnitude than the pooled Focal×Post coefficient in Table O1.1 (−1.62 in logs, an 80% decline for the same outcome). The export-value statistics for both events and the quantity statistic for October 2023 have p -values in the range [0.13, 0.23], consistent with the wide confidence bands visible in Figure O1.4. The raw fraction for the October 2022 quantity contrast is $1/54 \approx 0.019$.² The remaining outcomes’ larger p -values mean that those profiles are not comparably unusual relative to the comparison-unit assignments under this test; the corresponding event-study figures remain descriptive.

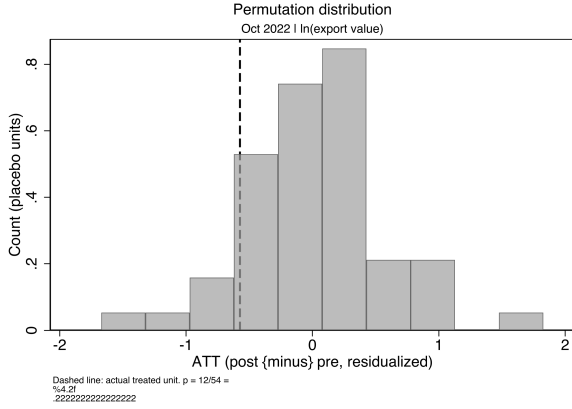
Table O2.1: Fisher permutation inference for the focal unit

Event	Outcome	Post-pre [†]	$N_{\text{comparison}}$	p -value
Oct 2022	ln(export value)	−0.574	54	0.222
Oct 2022	ln(quantity)	−1.088	54	0.019
Oct 2023	ln(export value)	+0.624	53	0.132
Oct 2023	ln(quantity)	+0.574	53	0.226

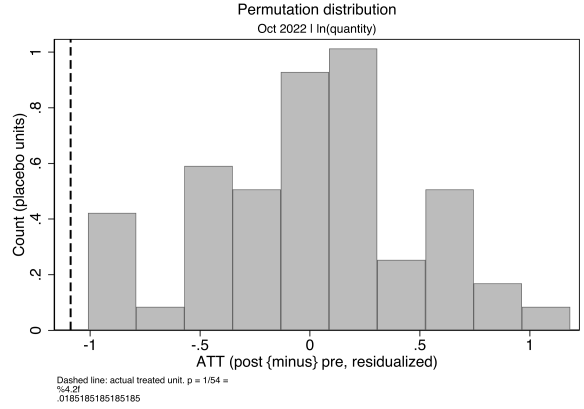
Notes: The focal unit is Taiwan×MCP-IC. Each remaining country×HS-code cell serves as a comparison assignment ($N_{\text{comparison}} \in \{53, 54\}$). Post-pre[†] is the mean residualized outcome in the post-event window ($k = 0, \dots, 12$) minus the pre-event window ($k = -12, \dots, -2$), after partialling out destination×month and item×month fixed effects and the AI demand control. The contrast is descriptive, not a causal effect. The p -value is the fraction of comparison-unit contrasts at least as large in absolute magnitude as the focal-unit contrast.

Figure O2.1 plots the comparison-unit distribution for each event–outcome combination. The dashed vertical line marks the Taiwan×MCP-IC statistic. The October 2022 quantity statistic lies in the far left tail; the other focal statistics lie nearer the centers of their distributions, consistent with Table O2.1.

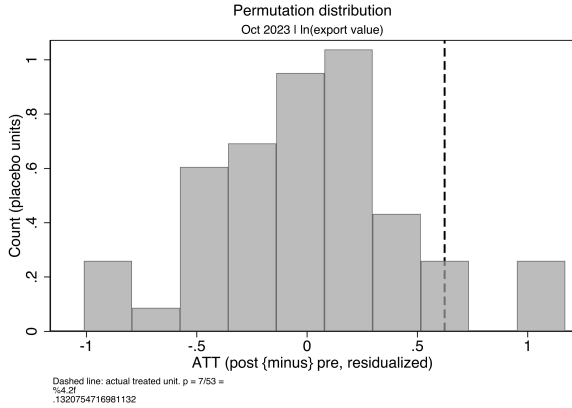
²This excludes the focal unit from the reference set; the more conservative randomization-inference convention $(1 + n)/(1 + N)$ gives $2/55 \approx 0.036$, which is likewise unusually extreme among the comparison assignments.



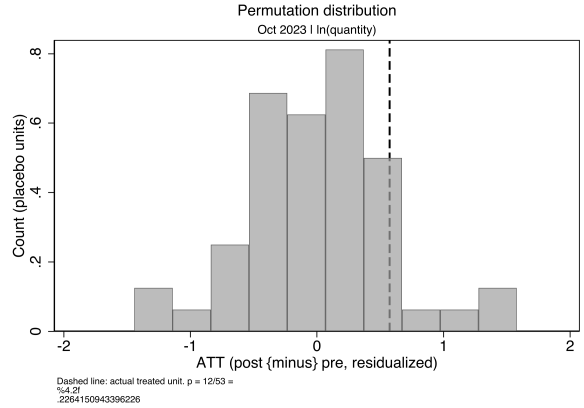
(a) Oct 2022: export-value statistic.



(b) Oct 2022: quantity statistic.



(c) Oct 2023: export-value statistic.



(d) Oct 2023: quantity statistic.

Figure O2.1: Comparison-unit distributions for the Fisher permutation test. Each panel shows the descriptive post-minus-pre statistic (labeled ATT on the panel axes) for the 53–54 comparison country $\times$ HS-code units. The dashed vertical line marks the Taiwan $\times$ MCP–IC statistic. The p -value is the fraction with $|\hat{C}_{\text{comparison}}| \geq |\hat{C}_{\text{focal}}|$; the contrasts are not interpreted as causal effects.

O3 Industry Details and Network Extensions

This section connects the AI/HBM industry evidence to the model’s network representation. Tables [O3.1](#) and [O3.2](#) document the production chain and concentration evidence; Table [O3.3](#) and the [bargaining-support](#) and [multinational-production](#) subsections then record the notation, bargaining support, and extension used in the quantitative exercises.

Industry Details

Table [O3.1](#) organizes the production ecosystem that motivates the model’s sparse buyer–supplier network. Panel A lists the three firms—SK hynix, Samsung, and Micron—that account for virtually the entire global supply of High-Bandwidth Memory (HBM), Panel B reports the leading accelerator designers and manufacturing partners, and Panel C summarizes segment concentration. NVIDIA is the primary buyer of HBM for data-center GPUs, with AMD and a limited set of cloud-native designers (e.g., Google, AWS) constituting the residual demand. The production network is highly globalized yet distinct from commodity memory chains: HBM production requires tight integration with advanced packaging capacity (e.g., TSMC’s Chip-on-Wafer-on-Substrate (CoWoS)), creating high switching costs and qualification barriers between buyers and suppliers.

Table [O3.2](#) separates the external upstream concentration evidence from the imposed downstream scenario proxy. Panel A reports external Q2 2025 HBM market shares from Counterpoint Research, chosen to reflect the quarter when Samsung had not yet cleared NVIDIA’s HBM3E qualification: SK hynix (62.0%), Samsung (17.0%), and Micron (21.0%).³ These shares imply an HHI of approximately 4,574, about 1.8 times the 2,500 threshold for a “highly concentrated” market under U.S. merger guidelines. Panel B reports a lower bound for downstream concentration: industry estimates place NVIDIA’s share of the AI accelerator market near 80%. The separate 90/10 HBM-exposure split used in calibration is an

³Corroborated by a second industry dataset (TrendForce, *2Q26 HBM Datasheet*). The market is projected to deconcentrate through 2026 as Samsung’s HBM4 ramps. The Q2-2025 split is the calibration snapshot; quantitative magnitudes may differ under other concentration structures.

Table O3.1: AI/HBM value chain: producers, manufacturing partners, and an indicative concentration snapshot

Panel A: HBM producers and production/assembly sites		
Chip maker	HQ (Country)	Major HBM fab (City, Country)
SK hynix	South Korea	Icheon, Cheongju (KR); Wuxi (CN)
Samsung Electronics	South Korea	Hwaseong, Pyeongtaek (KR); Xi'an (CN)
Micron Technology	USA	Taichung (TW, back-end); Hiroshima (JP)*
Panel B: AI accelerator designers and manufacturing/packaging partners		
Designer	Foundry (node)	Advanced packaging (type)
NVIDIA (H100/H200)	TSMC (5/4/3 nm)	TSMC CoWoS (2.5D with HBM)
AMD (MI300 series)	TSMC (5/4 nm)	TSMC CoWoS
Intel (Gaudi 3)	Intel/TSMC (var.)	EMIB/CoWoS/OSAT mix
Broadcom (custom XPU _s)	TSMC (advanced)	CoWoS/interposer
Panel C: Concentration snapshot (2024–2025, indicative)		
Segment	Recent concentration snapshot	
HBM suppliers	SK hynix, Samsung, Micron—three-firm oligopoly	
Advanced packaging	TSMC CoWoS is a key capacity bottleneck	
AI accelerators	NVIDIA is the dominant AI accelerator designer and HBM buyer	
HBM market size	Rapid growth from $\sim$ \$18B (2024) to $\sim$ \$34B (2025e)	

* Linked ecosystem/supply-chain site rather than a dedicated HBM fab.

Notes: Panel A sites reflect public reporting of representative DRAM/HBM front-end or back-end locations; specific qualified HBM3/3E lines are subsets of broader DRAM facilities. Panel B emphasizes that most leading accelerators rely on HBM and advanced packaging. Panel C entries summarize public trackers and company statements; Table O3.2 and its notes report the corresponding market-share sources and calibration role. In the model, I treat the upstream side as an HBM triopoly.

intentionally imposed high-concentration scenario proxy motivated by NVIDIA’s dominance; it is neither an observed moment, the broader accelerator-market share, nor direct bilateral HBM spending.

Table O3.2: External concentration snapshots for HBM suppliers and AI accelerators (indicative)

Panel A: HBM suppliers — market shares and concentration (Q2 2025)	
Supplier	Share (%)
SK hynix	62.0
Samsung	17.0
Micron	21.0
HHI	4,574
Panel B: AI accelerators — lower-bound concentration (2024–2025)	
Leader share (s_L , NVIDIA)	≥ 80
HHI lower bound	6,400

Notes: Panel A shares follow Counterpoint Research (Q2 2025), a quarter when Samsung had not yet qualified for NVIDIA’s HBM3E program; $HHI = 62.0^2 + 17.0^2 + 21.0^2 = 4,574$. Panel B reports a lower bound for accelerator concentration using the leader share $s_L \approx 80\%$; with only a leader identified, $CR1 \geq 80$ and $HHI_{\min} = s_L^2 = 6,400$. Panel A figures are used as calibration targets for upstream market shares in the static model.

Notation guide

Table O3.3 collects the firm indices, equilibrium objects, and structural parameters used in the model and quantitative protocols.

Bargaining Protocol and Model Support

This subsection states the fixed-link scope and passive-beliefs convention that support the bilateral bargaining system, and it records the factor-demand identity used to construct upstream values.

The static model omits dynamic and extensive margins: network and innovation dynamics (Lim, 2018; Oberfield, 2018; Klette and Kortum, 2004; Arkolakis et al., 2018), FDI

A. Indices and sets

D, U	sets of downstream and upstream firms
$d \in D, u \in U$	individual downstream / upstream firm
$\mathbf{q}, \mathbf{p}, \mathbf{C}$	generic vectors of quantities, prices, and costs

B. Variables (time index t implicit)

z_d, z_u	productivities (down-/up-stream)
p_d^D, q_d^D, C_d^D	downstream price, quantity, and unit delivery cost
$p_{du}^U, q_{du}^U, C_{du}^U$	input price, quantity, and unit delivery cost from u to d
P_d^M	materials (composite input) price index for downstream firm d ; the superscript M distinguishes the upstream input bundle from the downstream output price p_d^D

C. Parameters

σ, ρ, η	substitution elasticities
γ_d	bargaining weight of downstream firms

Table O3.3: Notation for firms, prices, quantities, costs, elasticities, and bargaining weights.

(Helpman et al., 2004; Ramondo and Rodríguez-Clare, 2013), inventories (Alessandria et al., 2010, 2011, 2013), relational contracting (Antràs and Helpman, 2004; Antràs and Foley, 2015), and capacity (Kreps and Scheinkman, 1983). Holding the active-link set fixed isolates intensive-margin input-price incidence. Bargaining is a standard way to model two parties in production networks that split the surplus generated by both contributions; Dhyne et al. (2023); Eaton et al. (2022); Lamadon et al. (2022) feature producers and factor suppliers dividing surplus by bargaining.

The CES cost map gives the factor-demand identity in equation (O3.1). By Shephard's lemma, the conditional demand for input u is

$$q_{du}^U = \frac{q_d^D}{z_d} \left(\frac{p_{du}^U}{P_d^M} \right)^{-\eta}. \quad (\text{O3.1})$$

Hence, the expenditure share on link (d, u) is $\omega_{du} = (p_{du}^U / P_d^M)^{1-\eta}$. Holding downstream output and all other link prices fixed, the conditional own-price elasticity is $-\eta(1 - \omega_{du})$ because $\partial \log P_d^M / \partial \log p_{du}^U = \omega_{du}$. The total equilibrium elasticity additionally includes the

induced response of q_d^D .

Definition 2 uses the passive-beliefs convention standard in bilateral contracting (Horn and Wolinsky, 1988).

Remark 2 (Passive vs. active beliefs). *Definition 2 adopts passive beliefs, standard in the empirical NiN literature because they yield a tractable fixed-point formulation and local comparative statics; Collard-Wexler et al. (2019) find passive and active beliefs deliver similar equilibrium prices. The maintained buyer-side log-submodularity condition is a passive-beliefs restriction. Cournot/CES demand determines the on-path profit component, but with a nonzero disagreement payoff the curvature of $\log \Delta \Pi_d^D$ also depends on the disagreement value and its response to the wedge. Active beliefs change those terms, so no protocol-robustness result is claimed. The solved directional comparative statics route through buyer and seller incremental surpluses and the signed entries of the passive-beliefs Jacobian (Remark 4).*

Extension to the Multi-Country Model

Adding Countries and Multinational Production

This subsection records an accounting extension for international trade and multinational production. The setup provides an accounting representation that can accommodate the Korea–Taiwan corridor in Section 4: Korean HBM fabs ($n(j) = \text{Korea}$) supply Taiwan accelerator plants ($n(i) = \text{Taiwan}$) through delivered plant–pair contract prices $p_{ij,du}^U$. The iceberg cost $\kappa_{n'i}$ instead applies downstream, from accelerator plant i to final destination n' . Consistent with the quantitative model, the extension retains a single HBM input class. A genuine multiple-input-type extension would require an additional aggregator and is not part of the accounting in Section O3. Table O3.4 fixes notation for sites and firms.

Countries and ownership Let $\mathcal{N}$ be countries, $\mathcal{D}$ downstream firms, and $\mathcal{U}$ upstream firms. Firm d operates plants $i \in \mathcal{I}_d$ with location map $n(i) \in \mathcal{N}$; similarly u operates plants $j \in \mathcal{J}_u$ with $n(j) \in \mathcal{N}$.

	Production stage	Production site	Firm
Input factors	upstream	j	u
AI accelerators	downstream	i	d
Output markets	final	n'	$\cdot$

Table O3.4: Site and firm indices for the multinational-production accounting.

Plant-specific costs and firm-level profits. For a buyer-plant i and a supplier-plant j , let $p_{ij,du}^U$ be the delivered contract price, inclusive of upstream j -to- i logistics and qualification frictions. The materials price index faced by plant i is

$$P_{i,d}^M = \left(\sum_{u \in \mathcal{U}} \sum_{j \in \mathcal{J}_u} (p_{ij,du}^U)^{1-\eta} \right)^{\frac{1}{1-\eta}}.$$

Let $\kappa_{n'i} > 0$ be the downstream iceberg cost from accelerator plant i to final destination n' . The unit cost (in plant- i currency) to serve n' is

$$C_{n',i,d}^{D,(n(i))} = \frac{\kappa_{n'i}}{z_d} P_{i,d}^M.$$

Profits are consolidated at the firm level. Firm d 's profit is

$$\Pi_d^D = \sum_{n'} \sum_{i \in \mathcal{I}_d} (p_{n',i,d}^D - C_{n',i,d}^D) q_{n',i,d}^D.$$

Bargaining at the plant-pair level. Each plant-pair price $p_{ij,du}^U$ is a separate Nash-in-Nash link. The bargain for (i, j, d, u) takes every other plant-pair price as given, including prices on other links between the same firm pair; it does not maximize a joint Nash product over a vector of plant-pair prices. Buyer and seller incremental surpluses are the changes in their respective firm-level values when that plant-pair link is active rather than severed, using the *plant-specific* cost map in Section O3.

The accounting and plant-pair bargaining definitions carry over, but Lemmas 1–2 do not by themselves establish uniqueness for an expanded multi-plant Cournot game. A general

extension requires a complete plant-destination demand and conduct system together with a certificate for the full plant-output Jacobian. I therefore condition on a selected interior Cournot branch with a single-valued, continuously differentiable mapping from plant-pair prices to quantities; no global multi-plant Cournot uniqueness result is claimed. The closed-form one-link seller-surplus calculation in Lemmas 3–4 carries over only under its stated flat-cost, fixed-local-elasticity and participation-endpoint restrictions.

The model delivers a tightly parameterized map from primitives to outcomes in the AI/HBM vertical. Given (i) the network of active links between HBM suppliers and accelerator designers, (ii) firm-level productivities $\mathbf{z}$, (iii) buyer-specific output wedges $\boldsymbol{\tau}$, and (iv) bargaining weights $\boldsymbol{\gamma}$, the numerical procedure selects a local price branch and its associated unit costs, quantities, and markups. Participation and pointwise regularity are maintained conditions for the theoretical comparative-static branch, not consequences of numerical price convergence. The multi-country accounting maps locations and trade frictions into the empirical setting in Section 4: HBM and accelerator plants can be located in different countries, plant-destination iceberg costs enter through $\kappa_{n'i}$, distinct from the firm-level policy wedge τ_d , and HBM scarcity at plant i is summarized by $P_{i,d}^M$. Section 3 studies how counterfactual “Chip War” shocks—changes in $\boldsymbol{\tau}$ —propagate through this structure, and Section 5 feeds calibrated AI/HBM primitives into it. The quantitative exercises trace model-predicted counterfactual incidence through HBM prices, rival buyers, and downstream markups; they do not recover observed bilateral incidence.

Parts (i)–(iii) record which elements of Propositions 1–2 carry over conditionally to the multinational accounting in Section O3.

(i) *Cournot block.* Firm d ’s profit aggregates plant-level revenues $\sum_{n'} \sum_{i \in \mathcal{I}_d} (p_{n',i,d}^D - C_{n',i,d}^D) q_{n',i,d}^D$, where each plant-specific unit cost $C_{n',i,d}^D = \kappa_{n'i} P_{i,d}^M / z_d$ inherits its CES materials index from the plant-pair links. Plantwise scalar curvature does not establish joint concavity or uniqueness when a firm’s plant-destination outputs interact. The single-output Lemmas 1–2 therefore do not supply a multi-plant theorem. Parts (ii) and (iii) are conditional on a

selected interior Cournot branch at which the full plant-output FOC Jacobian is nonsingular.

(ii) *One plant-pair link.* Fix all plant-pair prices except (i, j, d, u) . Buyer d 's incremental surplus $\Delta\Pi_d^D$ is the change in d 's Cournot profit when link (i, j) is active versus removed. Under the flat supplier cost, fixed local link elasticity, fixed other-link sales, and participation-endpoint conditions of Lemmas 3–4, those lemmas give the corresponding closed-form seller diagnostic and a unique interior one-link optimizer. In the convex-cost multi-link model, however, $\Delta\Pi_u^U$ is the difference in the seller's full firm profit across the with-link and without-link outcomes. No closed-form $p^\dagger$ or global one-link log-concavity claim is carried over to that case. The numerical routine uses the full surplus; the local or uniform Jacobian certificate in part (iii) is a separate maintained condition for regularity.

(iii) *Joint system.* Conditional on the selected single-valued C^1 Cournot mapping, stack all plant-pair FOCs into the pseudo-gradient g^{MC} , with Jacobian J^{MC} . If the multinational feasible box contains an interior zero and a fixed positive diagonal R^{MC} makes $\frac{1}{2}(R^{\text{MC}} J^{\text{MC}} + [J^{\text{MC}}]^\top R^{\text{MC}})$ negative definite throughout that box, the analogous conditional uniqueness result follows on that mapping. At a selected participation-satisfying multinational branch, pointwise negative definiteness of the same symmetric part instead yields nonsingularity, the P -matrix property, and local regularity. The comparative-statics derivative formula uses that local certificate; $-[J^{\text{MC}}]^{-1}$ has a strictly positive diagonal but need not be entrywise nonnegative, so signed responses require the specific inverse-Jacobian entries rather than a blanket sign claim.

O4 Computational and Calibration Protocols

Section O4 records the numerical algorithm used to compute the bilateral oligopoly equilibrium in Section 2. The target is a Nash-in-Nash upstream price vector associated with the link-level log-Nash objectives and downstream Cournot responses. The MATLAB implementation nests an inner Cournot solver, which maps unit costs into quantities and profits,

inside an outer NiN solver that updates link prices until the price-change stopping rule in step (iii) is met.

Inner loop: Cournot block

Given a candidate vector of upstream prices $\mathbf{p}^U$, the inner block computes downstream unit costs and the Cournot equilibrium defined in Definition 1.

(i) *materials price index and unit costs.* For each downstream firm d , the program computes the CES materials price index

$$P_d^M(\mathbf{p}^U) = \left(\sum_u (p_{du}^U)^{1-\eta} \right)^{\frac{1}{1-\eta}}$$

and the unit delivery cost

$$C_d^D(\mathbf{p}^U) = \frac{\tau_d}{z_d} P_d^M(\mathbf{p}^U).$$

These operations are vectorized over all active links (d, u) in the code.

(ii) *Cournot quantities and profits.* Taking $\mathbf{C}^D = (C_d^D)_d$ as given, the algorithm solves for the Cournot equilibrium quantities $\mathbf{q}^{D*}(\mathbf{C}^D)$ defined in Definition 1. The MATLAB code applies a simultaneous damped diagonal-Newton update in quantity levels. At each iteration, prices and marginal-profit residuals under (1) are evaluated at one common current quantity vector, a full candidate vector is formed, and all quantities are updated together. The Cournot block stops when the maximum KKT residual is below 10^{-10} or the maximum quantity change is below 10^{-12} , with at most 1,000 iterations.

Once convergence is reached, the code records, for each firm d , the equilibrium quantity q_d^{D*} , price p_d^D , and profit

$$\tilde{\Pi}_d^D(\mathbf{C}^D) = (p_d^D(\mathbf{q}^{D*}) - C_d^D) q_d^{D*},$$

as well as the link-level input demands q_{du}^U , supplier totals $Q_u = \sum_d q_{du}^U$, and upstream profits

$$\Pi_u^U(\mathbf{p}^U) = \sum_d (p_{du}^U - C_{du}^U) q_{du}^U(\mathbf{p}^U) - \frac{1}{2} c_{0u} \psi_u Q_u(\mathbf{p}^U)^2$$

using the CES demand system (O3.1). This is the full convex-cost profit in equation (10); the final term vanishes only in the flat-cost case. These objects are then passed to the outer NiN loop.

Outer loop: Nash-in-Nash over link prices

The outer loop treats the Cournot block as a black box that maps $\mathbf{p}^U$ into downstream and upstream surplus. It iteratively updates each link price to maximize the one-link log-Nash objective, holding all other prices fixed.

(i) *Incremental surplus on a link.* Fix a link $L = (d, u)$ and a current price vector $\mathbf{p}^U$. The code evaluates:

- the “with-link” outcome at $\mathbf{p}^U$ (all active links present);
- the “without-link” outcome at the same prices but with the (d, u) link severed numerically by setting $p_{du}^U = 10^9$, so that $\omega_{du} \approx 0$ and $q_{du}^U \approx 0$.

Each evaluation calls the inner Cournot block in Section O4 and returns the firm-level values Π_d^D and Π_u^U in the two scenarios. It recomputes every affected $q_{d'u}^U$ and Q_u before applying the full convex total cost. Thus the seller’s disagreement value includes changes in the supplier’s other-link sales and in $\frac{1}{2} c_{0u} \psi_u Q_u^2$, rather than treating seller surplus as link revenue net of a constant unit cost. The incremental surplus terms are then

$$\Delta \Pi_d^D = \Pi_d^D - \Pi_{d,-u}^D, \quad \Delta \Pi_u^U = \Pi_u^U - \Pi_{-d,u}^U.$$

These are functions of the single scalar p_{du}^U when all other link prices are held fixed.

(ii) *One-dimensional link update.* For link $L = (d, u)$, the algorithm considers the log-Nash objective

$$h_L(p_{du}^U) = \gamma_d \log \Delta \Pi_d^D + (1 - \gamma_d) \log \Delta \Pi_u^U,$$

with other link prices fixed at their current values. In the flat-cost, fixed-local-elasticity diagnostic of Lemmas 3–4, this objective is strictly concave on its stated feasible interval. The calibrated convex-cost model instead evaluates the full link objective h_L specified in Section O4; it does not invoke that diagnostic as a global concavity theorem. The link update searches for a maximizer of the full h_L on a bounded numerical interval.

The MATLAB code uses a safeguarded bounded golden-section search. At the start of each outer sweep, it fixes supplier volumes for the marginal-cost floors. For link (d, u) , this gives the lower bound p_{lo} and the initial bracket

$$p_0 = \max\{1.05p_{lo}, p_{du}^U\}, \quad p_{hi} = \max\{3p_0, p_0 + 1\}.$$

The safeguards then operate in order:

- (a) If the objective is nonfinite at p_0 , multiply p_0 by 1.5 for at most 10 retries. If all fail, set the link price to p_{lo} and skip the search.
- (b) If the objective is nonfinite at p_{hi} , multiply the upper bound by 1.8 for at most 10 retries. If it remains nonfinite, reset it to $\max\{5p_0, 10C_{du}^U + 1\}$.
- (c) If $p_0 \in (p_{lo}, p_{hi})$, symmetrically reseed the search interval around p_0 within those bounds.
- (d) Stop after at most 80 iterations or when $p_R - p_L < 10^{-10} \max\{1, |(p_R + p_L)/2|\}$, and return the best objective among the four final contracted points.

Incremental surpluses inside the log objective are floored numerically at 10^{-16} . These safeguards define the numerical link update; they are not a certificate of strict participation, candidate FOC satisfaction, or global one-link concavity.

(iii) *Cyclic best-response over links.* The full NiN solver uses Gauss–Seidel coordinate ascent

over links: it loops over buyers and then suppliers, skips inactive pairs, and refreshes the equilibrium snapshot immediately after each price update. In the sparse baseline, the order is NVIDIA–SK hynix, NVIDIA–Micron, AMD–Samsung, and AMD–Micron. The default initial price is $[\eta/(\eta-1)]C_{du}^U$. During each joint-calibration outer iteration, all share-objective evaluations use the preceding outer solve’s fixed p^U start. The algorithm:

- fixes all link prices except p_{du}^U ;
- runs the inner Cournot solver to compute the incremental surplus for (d, u) ;
- updates p_{du}^U to the maximizer of h_L as in step (ii);
- moves to the next link and repeats.

One full pass over all links produces an updated price vector $\mathbf{p}^{U,(1)}$. The procedure continues for $k = 0, 1, 2, \dots$ until the sup norm of the price change satisfies

$$\left\| \mathbf{p}^{U,(k+1)} - \mathbf{p}^{U,(k)} \right\|_{\infty} < \varepsilon_{\text{NiN}},$$

with $\varepsilon_{\text{NiN}} = 10^{-6}$ in absolute level prices and at most 3,000 outer iterations. This stopping rule is a numerical convergence diagnostic, not a theorem of global convergence or uniqueness.

Outputs and diagnostics

Once the price-change criterion is met, the algorithm performs a final Cournot solve to obtain the equilibrium quantities $\mathbf{q}^{D*}$, downstream prices $\mathbf{p}^{D*}$, and profits $\{\Pi_d^{D*}, \Pi_u^{U*}\}$. For the quantitative exercises in Section 5, the code also records:

- P_d^M and implied input-price component of downstream unit costs $\log P_d^M$;
- downstream markups and output prices;
- link-level expenditure shares ω_{du} and markups $p_{du}^{U*} - C_{du}^U$.

These objects construct the decompositions and incidence measures reported in Section 5. The reported calibrated runs meet the stated price-change criterion while the active link set is fixed.

Quantitative Protocols

Section O4 details the numerical algorithms and shock protocols used to generate the quantitative results in Section 5. The procedures are implemented in MATLAB and rely on the fixed-point solver in Section O4.

Full Calibration, Adjacency, and Fit

Table O4.1 expands the compact calibration summary in Section 5. The elasticities are imposed, literature-guided scenario values, the bargaining weights are imposed, and the active-link set is taken from the mid-2024 qualification network. Productivities are calibrated jointly to downstream accelerator-revenue and upstream HBM-revenue share ratios conditional on the externally anchored convex-cost stack.

The baseline values are scenario choices, not AI/HBM-specific estimates. I set $\sigma = 6$ guided by differentiated-product residual-demand ranges (Hottman et al., 2016; Feenstra and Weinstein, 2017); use $\rho = 4$ as a low-single-digit outer-nest scenario value (Atkeson and Burstein, 2008; Feenstra and Weinstein, 2017); and set $\eta = 2.5$ to represent incomplete but meaningful substitution across qualified HBM links, qualitatively motivated by evidence on input rigidity (Boehm et al., 2019; Oberfield and Raval, 2021). These sources do not identify AI/HBM-specific elasticities. Table O5.3 reports the feasible structural-elasticity grid. The separate +25% convex-cost stack remains the external operating-point anchor borrowed from Boehm and Pandalai-Nayar (2022).

The code’s two calibration statistics are

$$r_d^D := \frac{p_d^D q_d^D}{\sum_k p_k^D q_k^D}, \quad r_u^U := \frac{\sum_d p_{du}^U q_{du}^U}{\sum_v \sum_d p_{dv}^U q_{dv}^U}.$$

Table O4.1: Static calibration inputs, parameter roles, and the sparse baseline active-link set

Symbol	Object	Value	Role
σ	Downstream demand curvature	6.0	Cournot demand and markup response
ρ	Outer-nest substitution	4.0	Accelerator composite versus other final-good inputs
η	Within-bundle substitution	2.5	Substitution across qualified HBM links
D	Downstream firms	2	NVIDIA and AMD
U	Upstream firms	3	SK hynix, Samsung, and Micron
γ	Buyer bargaining weights	(0.5, 0.5)	Symmetric baseline; varied in robustness exercises
z_d, z_u	Relative productivities	Calibrated	Jointly match downstream and upstream revenue-share ratios
ζ	Convex-cost stack	+25%	External operating-point anchor
τ, f	Baseline wedges	1	Counterfactuals change only the stated buyer wedge

Buyer	Active HBM supplier	Baseline interpretation
NVIDIA	SK hynix	Qualified supplier
NVIDIA	Micron	Qualified supplier
AMD	Samsung	Qualified supplier
AMD	Micron	Qualified supplier

Notes: The Samsung–NVIDIA link is inactive in the sparse baseline and activated only in the extensive-margin robustness exercises.

Table O4.2: Imposed downstream scenario target, external upstream share targets, and calibrated model fit

Tier	Firm	Target share	Model share
Downstream revenue proxy	NVIDIA	0.90	0.90
Downstream revenue proxy	AMD	0.10	0.10
Upstream HBM revenue	SK hynix	0.62	0.62
Upstream HBM revenue	Samsung	0.17	0.17
Upstream HBM revenue	Micron	0.21	0.21

Notes: The 90/10 downstream target is an intentionally imposed high-concentration HBM-exposure scenario proxy motivated by NVIDIA’s dominance; it is neither an observed moment, the broader accelerator-market share, nor direct bilateral HBM spending. The 62/17/21 upstream targets are externally observed Q2-2025 Counterpoint HBM-revenue shares. Productivities are solved jointly at the externally borrowed +25% Nash-in-Nash convex-cost operating point. The maximum share residual is below 10^{-5} .

Thus r_d^D is a downstream accelerator-revenue share, distinct from the CES quantity share s_d in Lemma 1. The intentionally imposed 90/10 high-concentration HBM-exposure scenario proxy is mapped into r^D , while the externally observed Q2-2025 Counterpoint targets map directly into supplier HBM-revenue shares r^U . Let hats denote model-implied statistics. After normalizing AMD and Micron productivities, the inner calibration minimizes the three explicit log-ratio residuals,

$$\min_{z_d, z_u} \left\| \begin{bmatrix} \log(\hat{r}_{\text{NV}}^D / \hat{r}_{\text{AMD}}^D) - \log(0.90/0.10) \\ \log(\hat{r}_{\text{SK}}^U / \hat{r}_{\text{Micron}}^U) - \log(0.62/0.21) \\ \log(\hat{r}_{\text{Samsung}}^U / \hat{r}_{\text{Micron}}^U) - \log(0.17/0.21) \end{bmatrix} \right\|^2,$$

using a damped Levenberg–Marquardt step with backtracking. Each residual evaluation calls the equilibrium solver. An outer fixed point in the supplier cost shifters sets marginal cost to the common +25% stack at equilibrium volume. The stack remains an external anchor rather than a share-identified parameter. The calibration uses three independent share-moment ratios for the free productivity terms, but equality of moment and parameter counts is not a rank, identification, or uniqueness result.

Table O4.3: Joint productivity calibration procedure and numerical diagnostics

Step	Operation	Diagnostic
1	Normalize AMD and Micron productivities	One reference firm on each tier
2	Form one downstream and two upstream log revenue-share ratios	Two adding-up restrictions removed
3	Solve all free productivities jointly	Maximum share residual $< 10^{-5}$
4	Update supplier cost shifters	Common operating-point stack +25%
5	Resolve Nash-in-Nash equilibrium	Maximum link-price change $< 10^{-6}$

The fit is numerical discipline rather than an identification claim. Customs outcomes and the event-study path are not calibration targets. The price-change convergence diagnostic does not certify first-order conditions, participation, pointwise regularity, or global uniqueness; participation and pointwise regularity enter the theory as maintained conditions on the

selected branch.

Constructing the Frozen-Input Benchmark

To isolate the input-price incidence channel shown in Figure 2, I compare the full Nash-in-Nash equilibrium with a “frozen-input” benchmark. The benchmark holds negotiated link prices fixed, shutting down endogenous input repricing while downstream firms re-optimize quantities in response to the effective wedge. The algorithm proceeds as follows:

1. **Retrieve Baseline Prices.** Let $\mathbf{p}_0^{U*}$ denote the vector of negotiated link prices in the baseline equilibrium (Section 5), solved under baseline wedges $\boldsymbol{\tau}^0$. Let $\mathcal{L}$ be the set of active links.
2. **Apply Effective-Wedge Shock.** Introduce the counterfactual buyer-specific output wedges $\boldsymbol{\tau}^1$. For example, in the NVIDIA-exposed scenario, $\tau_{\text{NVIDIA}}^1 = 1.10 \times \tau_{\text{NVIDIA}}^0$.
3. **Compute Frozen Costs.** Calculate the new downstream unit costs $\mathbf{C}_{\text{frozen}}^D$ using the *shocked* wedges $\boldsymbol{\tau}^1$ but the *fixed* baseline link prices $\mathbf{p}_0^{U*}$. For each buyer d :

$$C_d^D(\text{frozen}) = \frac{\tau_d^1}{z_d} \left(\sum_{u \in \mathcal{U}_d} (p_{0,du}^{U*})^{1-\eta} \right)^{\frac{1}{1-\eta}}. \quad (\text{O4.1})$$

Crucially, this step bypasses the Nash-in-Nash update loop; firms calculate costs as if supplier contracts were rigid.

4. **Solve Partial Equilibrium.** Given the fixed cost vector $\mathbf{C}_{\text{frozen}}^D$, I solve for the new downstream Cournot quantities $\mathbf{q}^D$ and prices $\mathbf{p}^D$ using the algorithm in Definition 1.

The exact log identity in Equation (18) can be evaluated by comparing the full Nash-in-Nash outcome (where $\mathbf{p}^U$ adjusts to $\mathbf{p}_1^{U*}$) against this frozen benchmark. The input-price log change is zero by construction in the frozen benchmark. The difference in total downstream-price changes between the two equilibria also includes any difference in their

markup responses; Figure 2 reports the two endogenous components separately. The endogenous input-price and markup entries display log-percent changes, $100 \Delta \ln X$, while policy wedges are displayed in level percentage points. In particular, the displayed -3.56% and -0.12% are $100 \Delta \ln P_{NV}^M$ and $100 \Delta \ln \mu_{NV}$, so they are additive terms in the exact log identity after multiplying it by 100. Section 5 defines the reported 36% shorthand as a descriptive mixed-unit ratio rather than an exact common-scale decomposition.

The MC-Pricing (Zero-Margin) Benchmark

The three-pricing-rule comparison of Table O4.4 includes a benchmark intermediate between the frozen-input and full-Nash-in-Nash economies: every active link is priced at its link-specific supplier marginal cost, with zero upstream margin, while downstream firms retain Cournot competition. This is marginal-cost pass-through, not a Walrasian price-taking equilibrium; buyers keep their Cournot power. Under the convex stack, link (d, u) has marginal cost $MC_{du}(Q_u) = C_{du}^U + c_{0u}\psi_u Q_u$, where $Q_u = \sum_{d'} q_{d'u}^U$ (Section 2). Because Q_u depends on downstream quantities and hence on the prices the supplier sets, the benchmark is a fixed point in the link prices:

$$p_{du}^U = MC_{du}(Q_u(\mathbf{p}^U)) = C_{du}^U + c_{0u}\psi_u Q_u(\mathbf{p}^U), \quad (d, u) \in \mathcal{L}, \quad (\text{O4.2})$$

solved by damped iteration, recomputing Q_u from the downstream Cournot block at each step. That downstream block is the *same* solver used for the full-Nash-in-Nash and frozen legs, so the gap between any two legs reflects the pricing rule alone, not a change of numerical method.

The implementation changes the pricing rule without recalibrating productivities, cost floors, or curvature slopes: the ψ_u values anchored at the Nash-in-Nash +25% operating point are carried unchanged into the MC baseline. Because MC pricing changes equilibrium volumes, it does *not* re-create a common 25% stack. In supplier order (SK hynix, Samsung,

Micron), its baseline realizes

$$\psi_u Q_u = (0.312, 0.290, 0.404), \quad r^U = (0.615, 0.160, 0.225).$$

For (NVIDIA, AMD), the corresponding downstream accelerator-revenue shares are

$$r^D = (0.904, 0.096).$$

Two properties characterize the benchmark. First, on a flat supply curve ($\zeta = 0$) each link price collapses to its pair-specific constant unit cost C_{du}^U , invariant to the output wedge, so $100 \Delta \ln P_d^M = 0$ exactly—the correctness anchor. Second, each active supplier has non-negative total producer surplus at the fixed point. This is an accounting check, not the Nash-in-Nash link-level participation test: the MC-pricing benchmark has no negotiated agreement or severed-link disagreement point. Each pricing rule has its own baseline equilibrium. Accordingly, differences in reported log-percent responses are descriptive cross-rule gaps, not additive bargaining or supply components and not a structural surplus split. Comparability rests on all three rules reporting the same outcome $100 \Delta \ln P_d^M$, with Cournot competition retained downstream throughout. The exercise does not identify the convex stack; the external +25% anchor applies to the Nash-in-Nash calibration, while the MC baseline stacks reported with Table [O4.4](#) are implied outcomes.

Table O4.4: Input-price responses across pricing rules

	Frozen (fixed input)	MC-pricing (zero margin)	Full Nash-in-Nash	MC-Frozen gap (response gap)	NiN-MC gap (response gap)
<i>Scenario A</i> ($\tau_{NV}+10\%$, $\tau_{AMD}+5\%$), $100 \Delta \ln P^M$					
NVIDIA	+0.00	-4.46	-3.56	-4.46	+0.90
AMD	+0.00	-2.06	-1.55	-2.06	+0.51
<i>Scenario A output vector</i> , delivered instead by equal-bite buyer caps (NVIDIA $\sim 21\%$ cut), $100 \Delta \ln P^M$					
NVIDIA	+0.00	-9.39	+5.99	-9.39	+15.38
AMD	+0.00	-6.36	+3.66	-6.36	+10.02
Cross-buyer spillover: AMD's index under an NVIDIA-only +10% shock, $100 \Delta \ln P_{AMD}^M$					
AMD	+0.00	+0.51	+0.66	+0.51	+0.15

Notes: Entries are log-percent changes, $100 \Delta \ln P^M$. *Frozen* fixes upstream prices at their pre-shock Nash-in-Nash values. *MC-pricing* sets each active link to $MC_{du} = C_{du}^U + c_{0u}\psi_u Q_u$ with zero upstream margin and retains downstream Cournot competition. The cap panel removes both Scenario A wedges and caps each buyer at its own Scenario A wedge-equilibrium output. Each rule has its own baseline; the final two columns are differences between responses.

Heterogeneous Shock Protocols

The results in Figures 2 and O5.1 use two stylized asymmetric shock vectors to represent different “Chip War” compliance scenarios. These are defined for the counterfactuals as:

- **Scenario A (NVIDIA-exposed):** $\% \Delta \tau_{NVIDIA} = +10\%$, $\% \Delta \tau_{AMD} = +5\%$. This stylizes a regime where export controls bind more tightly on the dominant firm’s high-performance SKUs (e.g., H100/H200 restrictions).
- **Scenario B (AMD-exposed):** $\% \Delta \tau_{NVIDIA} = +5\%$, $\% \Delta \tau_{AMD} = +10\%$. This stylizes a regime where licenses or compliant product redesign mitigate NVIDIA’s constraint while the rival faces steeper effective barriers.

For the continuous sweep in Figure O5.3, I vary $\% \Delta \tau_{NVIDIA}$ from 5% to 50% in 5% increments while holding $\Delta \tau_{AMD} = 0$, solving the full Nash-in-Nash system at each step to trace the non-linear response of input prices.

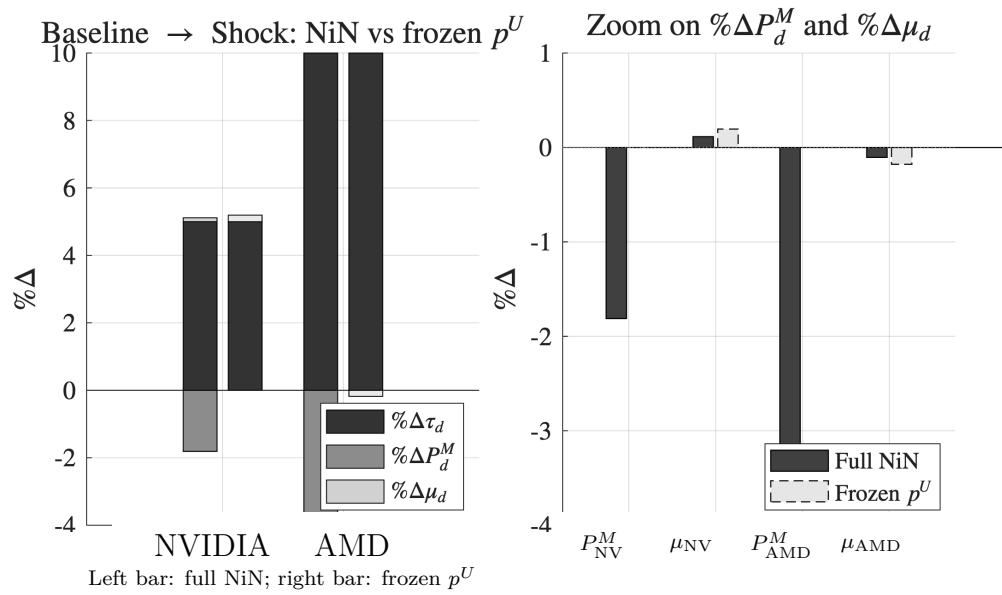


Figure O4.1: Reversed-exposure price decomposition (NVIDIA +5%, AMD +10%). Wedge bars are level percentage points; input-price and markup bars are log-percent changes, $100 \Delta \ln X$. Reversing exposure preserves the qualitative ordering: the more heavily restricted buyer receives the larger input-price offset, while markup responses remain much smaller.

O5 Quantitative Robustness and Policy Instruments

This section separates three robustness questions: whether network incidence changes when one link is added at fixed non-network primitives, whether the signs survive alternative bargaining weights and structural parameters, and whether a matched output cap changes the incidence boundary.

Fixed-Productivity Link Addition

Figure O5.1 compares the sparse baseline network with a dense network that adds the Samsung–NVIDIA link. Productivities, cost floors, bargaining weights, and curvature slopes are fixed across networks, and each economy is re-solved relative to its own pre-shock baseline.

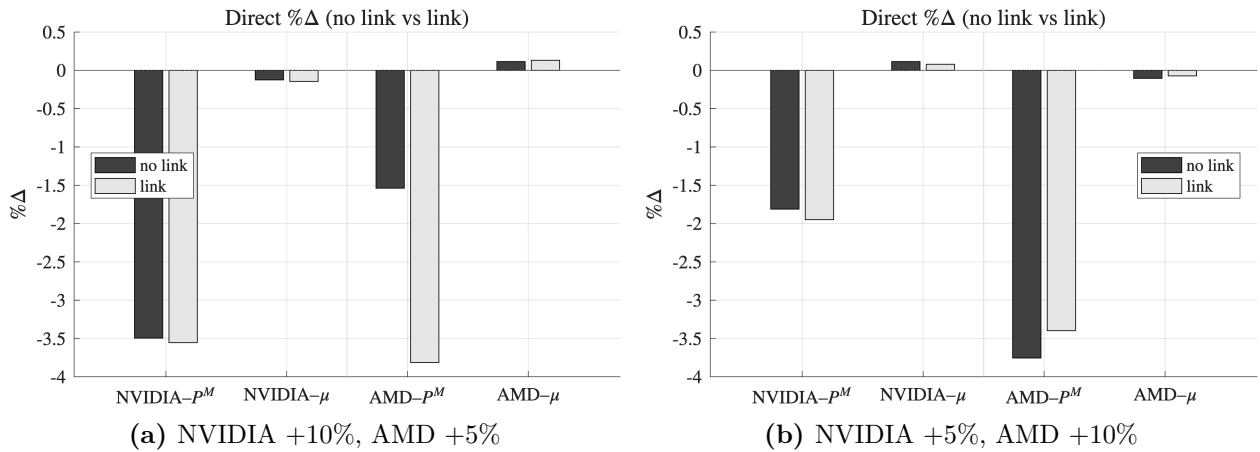


Figure O5.1: Fixed-primitive network non-neutrality with and without the Samsung–NVIDIA link. The sparse and dense economies hold all non-network primitives fixed and are solved separately. Each bar reports the ordinary percent change, $100(X_1/X_0 - 1)$, in the input-price index or markup in the full Nash-in-Nash equilibrium, relative to that economy’s own separately solved pre-shock baseline.

As discussed in Section 5, adding a Samsung–NVIDIA link changes NVIDIA’s baseline input bundle, bargaining options, input cost, and downstream accelerator-revenue share. This creates a potential confounding factor when comparing the incidence of trade shocks across the “link” and “no-link” economies: NVIDIA is a larger, lower-cost firm in the fixed-

productivity “with-link” counterfactual.

In a separate NVIDIA-only fixed-primitive experiment, I hold the canonical productivities, cost floors, bargaining weights, and ψ_u fixed, add only the Samsung–NVIDIA link, and solve a new dense baseline without share recalibration or stack re-anchoring. The dense baseline has $r^D = (0.946, 0.054)$ for (NVIDIA, AMD) and, in the order (SK hynix, Samsung, Micron), $r^U = (0.542, 0.301, 0.157)$ and $\psi_u Q_u = (0.258, 0.391, 0.228)$. Raising only τ_{NVIDIA} by 10% gives an AMD input-price-index response of -2.61% in ordinary percent (-2.64% in logs). Both the baseline and shock meet the solver’s price-change criterion. This cross-buyer diagnostic is distinct from the joint-shock own-incidence comparisons in Figure O5.1 and from the share-recalibrated comparison in Figure O5.2.

Figure O5.2 therefore reports a second comparison that restores both tiers’ baseline calibration statistics after activating the link. This procedure standardizes the pre-shock downstream accelerator-revenue and upstream HBM-revenue share vectors; it does not standardize every other equilibrium object.

Recalibration Algorithm

Let $\mathcal{E}_{\text{base}}$ be the calibrated baseline economy (no Samsung–NVIDIA link) and $\mathcal{E}_{\text{link}}$ be the counterfactual economy with the link active. The recalibration proceeds as follows:

1. **Target Baseline Statistics.** I retain $r_{\text{target}}^D = (0.90, 0.10)$ for (NVIDIA, AMD) and $r_{\text{target}}^U = (0.62, 0.17, 0.21)$ for (SK hynix, Samsung, Micron), using the revenue-share definitions in Section O4.
2. **Activate Link.** I add the $(d = \text{NVIDIA}, u = \text{Samsung})$ pair to the adjacency matrix for $\mathcal{E}_{\text{link}}$.
3. **Invert Jointly for Productivities.** Holding the canonical curvature slopes ψ_u fixed, I jointly recalibrate z'_d and z'_u , with AMD and Micron as the productivity normalizations, so that $\mathcal{E}_{\text{link}}$ matches both target vectors. The nonlinear least-squares problem

is

$$\min_{z'_d, z'_u} \left\| \begin{bmatrix} \log(\hat{r}_{NV}^D/\hat{r}_{AMD}^D) - \log(0.90/0.10) \\ \log(\hat{r}_{SK}^U/\hat{r}_{Micron}^U) - \log(0.62/0.21) \\ \log(\hat{r}_{Samsung}^U/\hat{r}_{Micron}^U) - \log(0.17/0.21) \end{bmatrix} \right\|^2, \quad (\text{O5.1})$$

solved by the same damped Levenberg–Marquardt routine as the baseline calibration. Because ψ_u is held fixed, this exercise does not re-anchor the common +25% stack after the link is added.

4. **Simulate Shocks.** The trade shocks defined in Section O4 are then applied to this recalibrated economy $\mathcal{E}'_{\text{link}}$.

Robustness of Incidence Results

Figure O5.2 isolates the part of the link comparison that remains after both tiers’ revenue-share targets are restored, while retaining the scope limits of the recalibration algorithm.

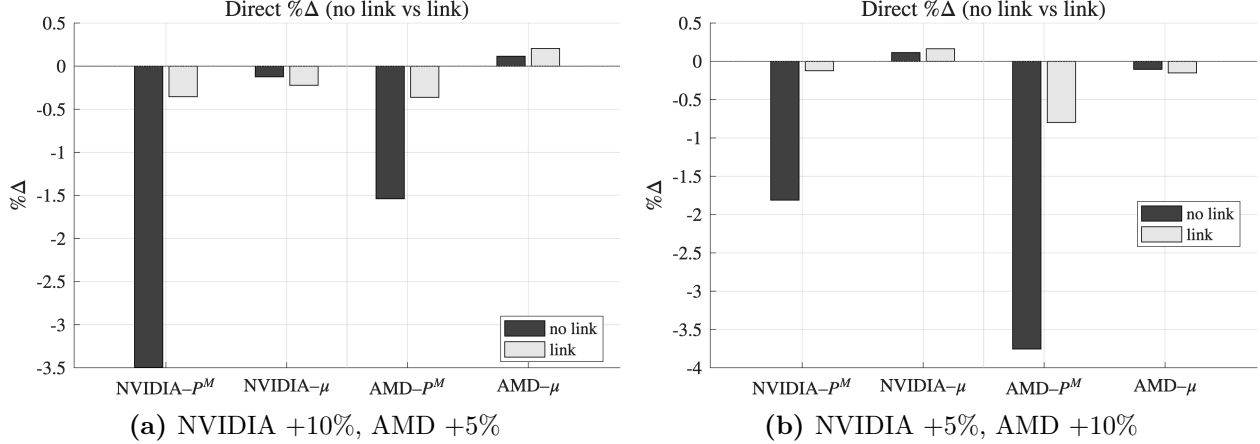


Figure O5.2: Share-recalibrated network non-neutrality, the counterpart to the fixed-primitive comparison in Figure O5.1. Downstream and upstream productivities are jointly recalibrated so that the link economy restores the pre-shock downstream accelerator-revenue and upstream HBM-revenue share targets. Each bar reports the ordinary percent change, $100(X_1/X_0 - 1)$, relative to the corresponding economy’s own separately calibrated pre-shock baseline.

The recalibration in Figure O5.2 offsets the new link’s effects on downstream scale and supplier revenue shares by changing relative productivities in both tiers, while leaving the canonical curvature slopes fixed.

Figure O5.2 shows substantial own-incidence non-neutrality after both revenue-share target vectors are restored. Adding the link attenuates NVIDIA's input-index decline from roughly -3.5% to -0.4% under NVIDIA-heavy exposure and from roughly -1.8% to -0.1% under reversed exposure; the markup bars change as well. Both buyers receive wedges in each panel, so this figure is not an NVIDIA-only cross-buyer spillover exercise. It shows that own incidence remains sensitive to the added link after share standardization. It does not isolate topology from every level difference: productivities, input costs, markups, volumes, link prices, and realized stacks may still differ across the two economies.

Wedge and Bargaining-Weight Sensitivity

Figures O5.3 and O5.4 vary the shock size and NVIDIA's bargaining weight; Tables O5.1, O5.3, and O5.4 report the discrete bargaining, structural-elasticity, and convex-stack checks.

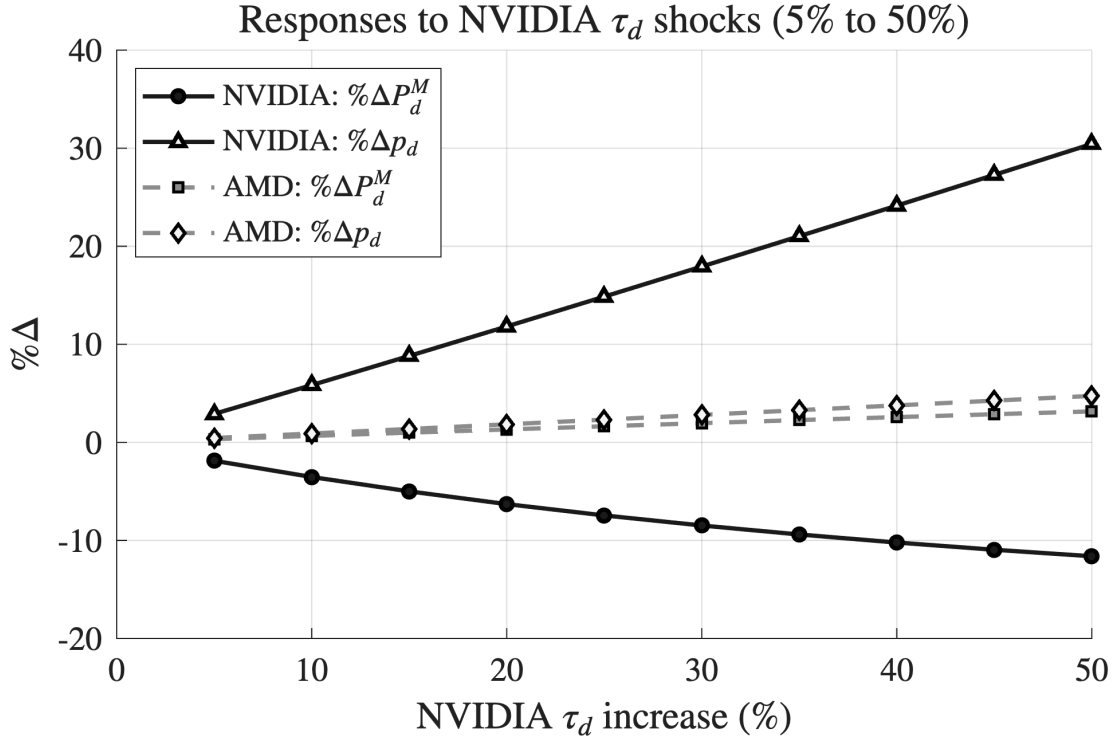


Figure O5.3: Responses of accelerator input costs and prices to NVIDIA's trade wedge. NVIDIA's wedge ranges from 5 to 50 percentage points above its baseline level while AMD's remains at baseline; plotted outcomes are ordinary percent changes, $100(X_1/X_0 - 1)$, from the pre-shock equilibrium for $X \in \{P_d^M, p_d^D\}$.

Figure O5.3 traces the nonlinear response to an NVIDIA-only wedge. NVIDIA’s input-price index falls as its wedge expands, while AMD’s index rises at the sparse calibration because AMD expands toward its non-shared supplier. The downstream price response remains dominated by the direct wedge, but endogenous input repricing grows with the shock. The index signs in Figure O5.3 and link signs in Figure O5.4 are calibration-verified responses for the specified network, not analytic sign restrictions.

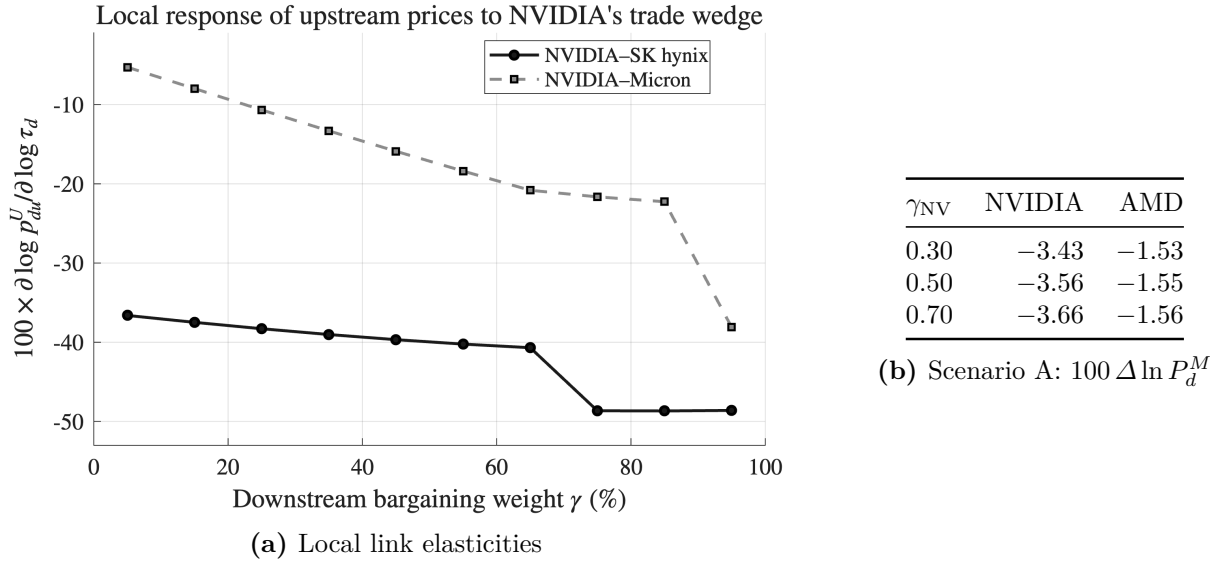


Figure O5.4: Sensitivity to NVIDIA’s bargaining weight. Panel (a) reports calibration-verified link-price elasticities; panel (b) reports Scenario A buyer-index log-percent changes, $100 \Delta \ln P_d^M$. Table O5.1 gives both asymmetric scenarios and markup responses.

Figure O5.4 separates link-level from buyer-level sensitivity. Panel (a) varies γ_{NV} from 0.05 to 0.95. Panel (b) reports buyer-level index responses at 0.30, 0.50, and 0.70. Higher buyer weight changes magnitudes, but the targeted buyer’s input-price response remains negative under convex cost in both asymmetric scenarios.

Remark 3 (Robustness of the absorber sign to buyer power). *Under flat upstream cost, the local derivative $\partial \log p_{NV,u}^{U*} / \partial \log \tau_{NV}$ turns positive near $\gamma_{NV} \approx 0.85$: at very high buyer power, the supplier’s margin motive can dominate its volume motive. The convex-cost calibration removes that boundary over the reported range because a contraction lowers marginal cost on the remaining business and reinforces the price cut. This is a conditional numerical*

comparison, not a global sign theorem.

Table O5.1: Sensitivity of incidence to bargaining weight γ_{NV}

γ_{NV}	Scenario A: NV +10%, AMD +5%		Reversed exposure: NV +5%, AMD +10%	
	$100 \Delta \ln P_{NV}^M / 100 \Delta \ln \mu_{NV}$	$100 \Delta \ln P_{AMD}^M / 100 \Delta \ln \mu_{AMD}$	$100 \Delta \ln P_{NV}^M / 100 \Delta \ln \mu_{NV}$	$100 \Delta \ln P_{AMD}^M / 100 \Delta \ln \mu_{AMD}$
0.30	-3.43% / -0.15%	-1.53% / 0.14%	-1.76% / 0.13%	-3.86% / -0.12%
0.50	-3.56% / -0.12%	-1.55% / 0.11%	-1.83% / 0.11%	-3.83% / -0.11%
0.70	-3.66% / -0.10%	-1.56% / 0.09%	-1.89% / 0.10%	-3.78% / -0.09%

Notes: Each cell reports the log-percent changes $100 \Delta \ln P_d^M$ and $100 \Delta \ln \mu_d$ in the materials price index and the Cournot markup $\mu_d = p_d^D / C_d^D$ relative to the baseline at the corresponding γ_{NV} . Across bargaining weights the input-price channel dominates the markup adjustment for the more heavily restricted buyer. The baseline calibration uses $\gamma_{NV} = \gamma_{AMD} = 0.5$; γ_{AMD} is held at 0.5 throughout.

Table O5.1 gives the same sign on the discrete bargaining-weight grid for both asymmetric shock scenarios. To assess whether the symmetric baseline itself matters, I also recalibrate productivities under $\gamma_{NV} = 0.7$ and $\gamma_{AMD} = 0.3$.

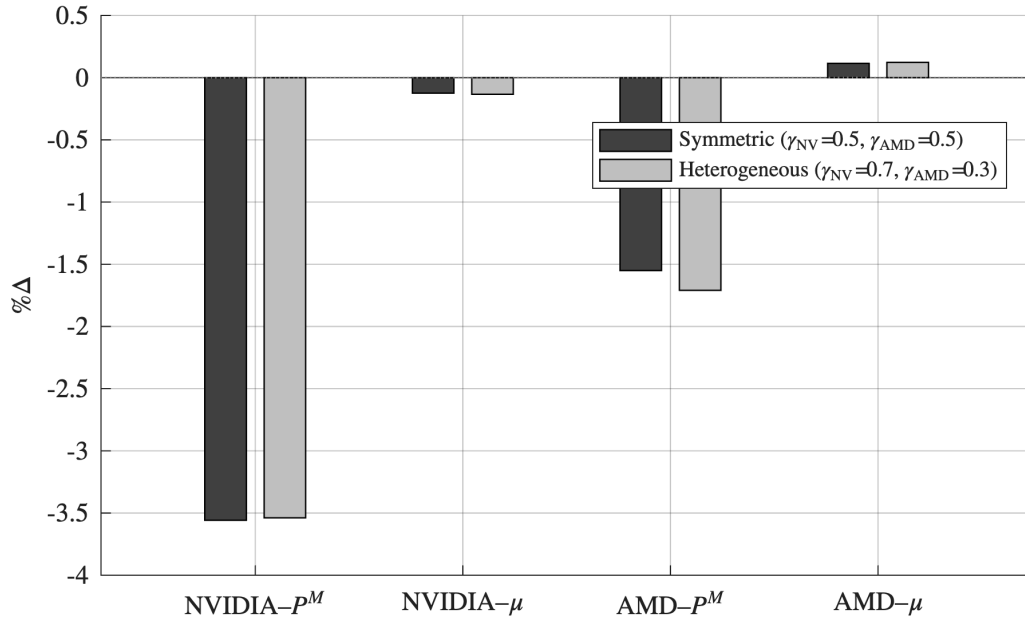


Figure O5.5: Scenario A log-percent decomposition, $100 \Delta \ln P_d^M$ and $100 \Delta \ln \mu_d$, under symmetric and heterogeneous bargaining weights. Productivities are recalibrated separately.

Table O5.2: Price decomposition under symmetric vs. heterogeneous bargaining weights (Scenario A: $\tau_{\text{NV}} + 10\%$, $\tau_{\text{AMD}} + 5\%$)

Configuration	$100 \Delta \ln P_{\text{NV}}^M$	$100 \Delta \ln \mu_{\text{NV}}$	$100 \Delta \ln P_{\text{AMD}}^M$	$100 \Delta \ln \mu_{\text{AMD}}$
Symmetric ($\gamma_{\text{NV}} = \gamma_{\text{AMD}} = 0.5$)	-3.559%	-0.125%	-1.551%	+0.115%
Heterogeneous ($\gamma_{\text{NV}} = 0.7$, $\gamma_{\text{AMD}} = 0.3$)	-3.538%	-0.133%	-1.709%	+0.123%

Notes: Each row reports the log-percent changes $100 \Delta \ln P_d^M$ and $100 \Delta \ln \mu_d$ in the materials price index and the Cournot markup μ_d for NVIDIA and AMD under Scenario A ($\tau_{\text{NV}} + 10\%$, $\tau_{\text{AMD}} + 5\%$). Downstream and upstream productivities are re-calibrated jointly for each γ configuration to maintain the baseline accelerator-revenue shares r^D and HBM-revenue shares r^U , with ψ_u held fixed. Signs and qualitative patterns are unchanged across configurations.

The targeted buyer’s input-price log-percent response is nearly unchanged under the heterogeneous calibration (-3.56% at symmetric weights and -3.54% at $\gamma_{\text{NV}} = 0.7$), while markup responses remain much smaller. Tables O5.3 and O5.4 report the structural-elasticity and convex-stack sensitivity.

Table O5.3 reports log-percent responses, $100 \Delta \ln P_d^M$ and $100 \Delta \ln \mu_d$. At each grid point, downstream and upstream productivities are jointly recalibrated to the baseline revenue-share targets with the canonical ψ_u fixed and no +25% stack re-anchoring. In supplier order (SK hynix, Samsung, Micron), the realized baseline $\psi_u Q_u$ vectors are

σ	$\eta = 2.5$	$\eta = 4.0$
4	(0.2756, 0.1998, 0.2442)	(0.0070, 0.0024, 0.0136)
6	(0.2481, 0.2481, 0.2481)	(0.0036, 0.0017, 0.0093)
8	(0.2422, 0.2774, 0.2541)	(0.0028, 0.0015, 0.0082)

At the disciplined within-bundle elasticity $\eta = 2.5$, the input-price response is stable across $\sigma \in \{4, 6, 8\}$. The smaller responses at $\eta = 4$ jointly reflect productivity recalibration and sharply different realized stacks, so the cross-row comparison does not identify an η -only effect. The $\eta = 1.5$ candidates have negative total supplier profit and fail a separate reporting screen. No shock responses are reported for them; this screen is not the link-level Nash-in-Nash participation test, and no sign result is inferred.

Table O5.3: Sensitivity of incidence to structural elasticities

η	σ	Panel A: NVIDIA ($100 \Delta \ln P_{NV}^M / 100 \Delta \ln \mu_{NV}$)			Panel B: AMD ($100 \Delta \ln P_{AMD}^M / 100 \Delta \ln \mu_{AMD}$)		
		$\sigma = 4$	$\sigma = 6$	$\sigma = 8$	$\sigma = 4$	$\sigma = 6$	$\sigma = 8$
1.5		n.r.	n.r.	n.r.	n.r.	n.r.	n.r.
2.5		-3.742% / -0.000%	-3.559% / -0.125%	-3.533% / -0.222%	-1.786% / -0.000%	-1.551% / +0.115%	-1.337% / +0.197%
4.0		-0.171% / -0.000%	-0.176% / -0.214%	-0.247% / -0.391%	-0.120% / -0.000%	+0.064% / +0.197%	+0.222% / +0.347%

Notes: Each cell reports $100 \Delta \ln P_d^M$ and $100 \Delta \ln \mu_d$ under Scenario A, with $\rho = 4.0$. At each feasible (σ, η) point, productivities are jointly recalibrated to $r^D = (0.90, 0.10)$ and $r^U = (0.62, 0.17, 0.21)$ while the canonical ψ_u is fixed and the +25% stack is not re-anchored. **Bold** marks the baseline $(\sigma, \eta) = (6, 2.5)$; “n.r.” denotes an omitted candidate under the nonnegative-total-profit reporting screen.

Table O5.4: Sensitivity of input-price absorption to the convex marginal-cost stack ζ

Stack $\zeta = \psi_u Q_u^*$	Implied $e = \frac{\zeta}{1+\zeta}$	$100 \Delta \ln P_{NV}^M$	$100 \Delta \ln \mu_{NV}$	Mixed-unit ratio
0%	0.00	-0.09%	-0.21%	1%
10%	0.09	-1.97%	-0.16%	20%
25%	0.20	-3.56%	-0.12%	36%
50%	0.33	n.r. (screened out)		

Notes: Each row jointly recalibrates (z_d, z_u) to the baseline revenue-share targets at the indicated ζ , then reports Scenario A log-percent changes. The final column uses the mixed-unit ratio defined in Section 5; $e = \zeta/(1 + \zeta)$ is the implied inverse supply elasticity.

Table O5.4 shows a convex-stack schedule ranging from negligible absorption under flat cost to the 36% descriptive ratio defined in Section 5 at the calibrated +25% point. At the operating point, $\zeta = \psi_u Q_u^*$ scales the convex marginal-cost addition relative to each supplier’s active-link-average floor c_{0u} ; it is not a common percentage markup over every pair-specific floor C_{du}^U . The mapping $e = \zeta/(1 + \zeta)$ makes the calibrated +25% stack an inverse supply elasticity of 0.20, between the low-utilization 0.12 and mean 0.27 estimates in Boehm and Pandalai-Nayar (2022).

The targeted buyer’s input-price response is negative at the reported 0%, 10%, and 25% stack points. The 50% candidate has negative total supplier profit and is omitted under the separate reporting rule; this is not the link-level Nash-in-Nash participation test, and the candidate is not treated as an equilibrium or sign result.

Alternative Policy Instrument

The cost path scales the two Scenario A wedges in their 2:1 ratio, leaving both buyers' quantities responsive to negotiated input prices. At each point, the equal-bite cap path removes both wedges and instead caps each buyer at its own equilibrium output on the cost path. This matched cap is a constructed instrument comparison, not an observed policy. Figure O5.6 and Table O5.5 report the resulting comparison against NVIDIA's output reduction; at the headline point NVIDIA's ceiling is about 21% below baseline and AMD is capped at its corresponding Scenario A output.

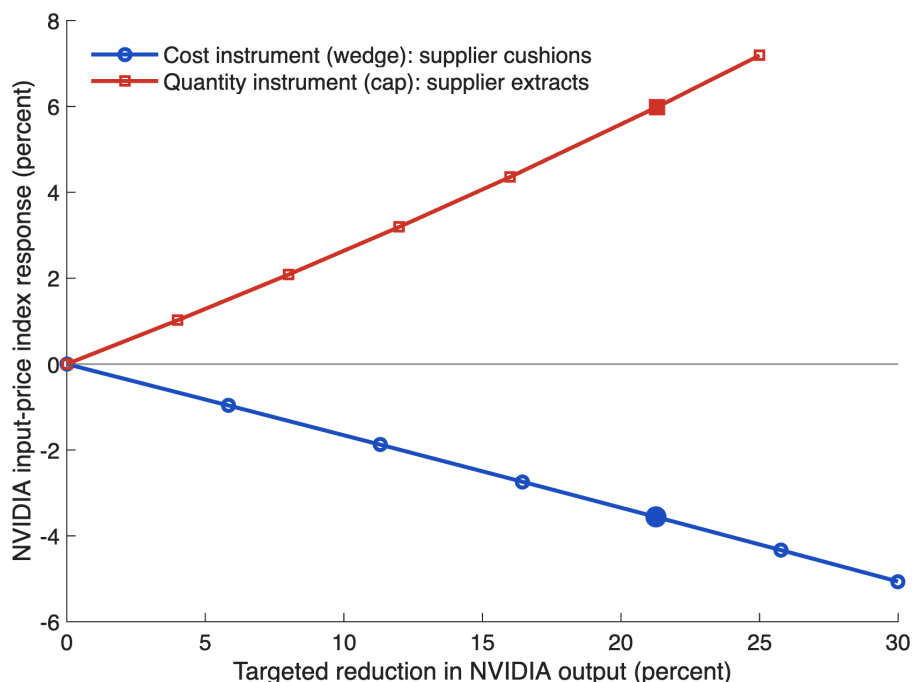


Figure O5.6: NVIDIA's log-percent input-price response, $100 \Delta \ln P_{NV}^M$, under the Scenario A wedge and constructed equal-bite two-buyer cap vectors. Filled markers denote the headline 21% output reduction; the caps are matched counterfactuals, not observed policy.

Table O5.5: Input-price incidence depends on the policy instrument

Targeted NVIDIA output cut	Cost instrument (wedge) $100 \Delta \ln P_{NV}^M$	Quantity instrument (cap) $100 \Delta \ln P_{NV}^M$
8%	−0.96%	+2.08%
12%	−1.88%	+3.20%
21% (headline)	−3.56%	+5.99%
25%	−4.33%	+7.19%

Notes: The response columns report $100 \Delta \ln P_{NV}^M$. The cost path scales the two Scenario A wedges in their 2:1 ratio. At each point, the equal-bite cap path removes both wedges and caps each buyer at its own cost-path output. The first column reports NVIDIA’s output cut; the table reports own-input-price responses, not cross-buyer effects.

In the calibrated comparison, NVIDIA’s input-price index falls about 3.6% in log-percent terms under the cost wedge and rises about 6.0% in log-percent terms under the equal-bite cap. The marginal-cost-pricing cap leg still falls about 9.4% in log-percent terms along the convex supply curve. This computed sign reversal is an equal-bite positive-margin boundary rather than a Nash-in-Nash-specific contribution: monopoly pricing to a captive buyer could reverse the sign as well, while bargaining determines the size of the margin response. No general instrument-sign theorem is claimed. The constrained Cournot and Nash-in-Nash system is derived in Section O9.

O6 Proof of Proposition 1

Proof roadmap.

Proof sketch. The proof has three steps. First, CES demand curvature in Lemma 1 and $\sigma \leq 2\rho$ yield strict row diagonal dominance of the elasticity-scaled Cournot marginal-profit Jacobian in Lemma 2. Second, Lemmas 3–4 establish one-link strict concavity only for the flat-cost, fixed-local-elasticity diagnostic; the convex-cost computation instead uses full seller profit. Third, Lemma 5(b) gives global uniqueness only under fixed-metric uniform negative definiteness. Under the maintained participation and pointwise regularity conditions,

Lemma 5(c) supplies negative own curvature, an invertible Jacobian, and local uniqueness at the selected convex-cost equilibrium. Lemmas 1–5 provide the details. $\square$

Under the uniform condition in Lemma 5(b), the result pins down a unique vector of negotiated link prices $\mathbf{p}^{U*}$ and, through the cost map and Cournot block, a unique downstream quantity profile. The pointwise condition in Lemma 5(c) supports only the local branch through the computed equilibrium.

Convex upstream cost. For $\psi_u > 0$, Assumption 2(c) maintains positive incremental surpluses and $S_R(\mathbf{p}^{U*}) \prec 0$ on the selected continuation branch. Lemma 5(c) therefore gives an invertible J , the P -matrix property, and local uniqueness along that branch; it does not establish negative definiteness elsewhere in $\mathcal{P}$. No fixed-metric analytic certificate of global uniqueness is available under convex cost.

The comparative-statics proof tracks the relevant entries of $-J^{-1}$; Remark 4 records why the maintained certificate alone does not sign them. Remark 5 gives only an accounting identity between the flat- and convex-cost derivatives, not a low-rank structural decomposition. The deflationary own-response and cross-buyer spillover remain network-conditional, calibration-verified statements.

Lemma 1 establishes the demand-curvature objects used in the markup formulas and in the strict-row-diagonal-dominance argument of Lemma 2.

Lemma 1 (Demand curvature and perceived elasticity). *Under (1),*

$$p_d^D = A \left(\frac{q_d^D}{Q^D} \right)^{-1/\sigma} (Q^D)^{-1/\rho}, \quad Q^D = \left(\sum_k (q_k^D)^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}, \quad \sigma > 1, \quad \rho \in (1, \sigma],$$

let

$$s_d = \frac{(q_d^D)^{(\sigma-1)/\sigma}}{\sum_k (q_k^D)^{(\sigma-1)/\sigma}} \in (0, 1), \quad \epsilon_d = \left(\frac{s_d}{\rho} + \frac{1-s_d}{\sigma} \right)^{-1}.$$

Then, holding $\mathbf{q}_{-d}$ fixed:

$$(a) \text{ **Perceived elasticity.** } \quad \frac{\partial \log p_d^D}{\partial \log q_d^D} = -\frac{1}{\epsilon_d} = -\left[\frac{1-s_d}{\sigma} + \frac{s_d}{\rho} \right].$$

(b) **Marginal revenue.** $MR_d = p_d^D \left(1 - \frac{1}{\epsilon_d}\right)$ and $\partial MR_d / \partial q_d^D < 0$.

(c) **Markup FOC.** With constant unit cost $C_d^D > 0$,

$$p_d^D = \frac{\epsilon_d}{\epsilon_d - 1} C_d^D.$$

Proof. Write $\log p_d^D = \log A - \frac{1}{\sigma} \log q_d^D + \left(\frac{1}{\sigma} - \frac{1}{\rho}\right) \log Q^D$. Let $M = \sum_k (q_k^D)^{\frac{\sigma-1}{\sigma}}$, so $\log Q^D = \frac{\sigma}{\sigma-1} \log M$ and

$$\frac{\partial \log M}{\partial \log q_d^D} = \frac{q_d^D \partial M / \partial q_d^D}{M} = \frac{\sigma-1}{\sigma} \frac{(q_d^D)^{\frac{\sigma-1}{\sigma}}}{M} = \frac{\sigma-1}{\sigma} s_d.$$

Hence $\frac{\partial \log Q^D}{\partial \log q_d^D} = s_d$ and

$$\frac{\partial \log p_d^D}{\partial \log q_d^D} = -\frac{1}{\sigma} + \left(\frac{1}{\sigma} - \frac{1}{\rho}\right) s_d = -\left[\frac{1-s_d}{\sigma} + \frac{s_d}{\rho}\right] = -\frac{1}{\epsilon_d},$$

which proves (i).

For (ii), note that $\partial p_d^D / \partial q_d^D = (p_d^D / q_d^D) \partial \log p_d^D / \partial \log q_d^D = -(p_d^D / q_d^D)(1/\epsilon_d)$. Thus $MR_d = p_d^D + q_d^D \partial p_d^D / \partial q_d^D = p_d^D(1 - 1/\epsilon_d)$. Differentiate w.r.t. q_d^D (holding $\mathbf{q}_{-d}$ fixed):

$$\frac{\partial MR_d}{\partial q_d^D} = \underbrace{\frac{\partial p_d^D}{\partial q_d^D}}_{<0} \left(1 - \frac{1}{\epsilon_d}\right) - p_d^D \frac{\partial}{\partial q_d^D} \left(\frac{1}{\epsilon_d}\right).$$

Since s_d increases with q_d^D and $\frac{\partial}{\partial s_d}(1/\epsilon_d) = (1/\rho - 1/\sigma) \geq 0$ (by $\rho \leq \sigma$), $\frac{\partial}{\partial q_d^D}(1/\epsilon_d) \geq 0$. The first term is strictly negative (downward-sloping inverse demand with $\epsilon_d > 1$), and the second term is nonpositive, so $\partial MR_d / \partial q_d^D < 0$.

For (iii), with constant unit cost C_d^D , the Cournot FOC sets $MR_d = C_d^D$. Using the expression in (ii), $p_d^D(1 - 1/\epsilon_d) = C_d^D$, which rearranges to the stated markup formula. $\square$

Lemma 2 builds on Lemma 1 to establish existence and uniqueness of the Cournot equilibrium. The restriction $\sigma \leq 2\rho$ supplies a separate Jacobian-regularity certificate.

Lemma 2 (Cournot equilibrium: uniqueness and regularity). *Let $\sigma > 1$ and $\rho \in (1, \sigma]$. With constant unit costs $C_d^D > 0$, the Cournot game admits a unique equilibrium. If*

$$\sigma \leq 2\rho,$$

the elasticity-scaled marginal-profit Jacobian is strictly row diagonally dominant with positive diagonal; hence $-J^C$ is a P-matrix and J^C is nonsingular.

Proof. Write

$$\beta \equiv \frac{\sigma - 1}{\sigma}, \quad \varphi \equiv \frac{1}{\rho} - \frac{1}{\sigma} \geq 0, \quad \Lambda \equiv A Q^{-1/\rho}, \quad f(s) \equiv s^{-1/(\sigma-1)}(\beta - \varphi s).$$

Because $q_d = s_d^{\sigma/(\sigma-1)} Q$, the Cournot first-order conditions are

$$C_d^D = \Lambda f(s_d).$$

On $s \in (0, 1)$, $\beta - \varphi s > 0$ and

$$\frac{d \log f(s)}{ds} = -\frac{1}{(\sigma - 1)s} - \frac{\varphi}{\beta - \varphi s} < 0.$$

Thus f is continuous and strictly decreasing, with $f(s) \rightarrow +\infty$ as $s \downarrow 0$ and $f(1) > 0$. Set $\bar{\Lambda} = \min_i C_i^D / f(1)$. For $\Lambda \in (0, \bar{\Lambda})$, define $s_i(\Lambda) = f^{-1}(C_i^D / \Lambda)$. Each $s_i(\Lambda)$ is continuous and strictly increasing in Λ . Their sum converges to zero as $\Lambda \downarrow 0$. As $\Lambda \uparrow \bar{\Lambda}$, a firm attaining the minimum has share tending to one while every other share is positive, so the sum rises above one. Hence there is a unique Λ for which $\sum_i s_i(\Lambda) = 1$. It follows that

$$Q = \left(\frac{A}{\Lambda} \right)^\rho, \quad q_i = s_i^{\sigma/(\sigma-1)} Q$$

are uniquely determined. Lemma 1 gives strict own concavity, so these first-order conditions are optimal. This proves existence and uniqueness directly in share space.

For the local regularity statement, let J^C be the Cournot Jacobian of the marginal-profit map $F_d(\mathbf{q}) = MR_d - C_d^D$, with entries $J_{dk}^C = \partial MR_d / \partial q_k$. Write $\theta_d \equiv 1/\epsilon_d = \frac{1-s_d}{\sigma} + \frac{s_d}{\rho}$. Differentiating $MR_d = p_d(1 - \theta_d)$ (Lemma 1), using $\partial \log Q / \partial \log q_k = s_k$ and $\partial s_d / \partial \log q_k = \beta s_d(\delta_{dk} - s_k)$, gives the elasticity-scaled Jacobian entries

$$-\frac{q_d}{p_d} J_{dd}^C = \theta_d(1 - \theta_d) + \varphi \beta s_d(1 - s_d) > 0, \quad -\frac{q_k}{p_d} J_{dk}^C = \varphi s_k[(1 - \theta_d) - \beta s_d] \quad (k \neq d).$$

Collect these terms in $\tilde{A}$, where $\tilde{A}_{dk} = -(q_k/p_d)J_{dk}^C$. Then

$$-J^C = \text{diag}(p_d) \tilde{A} \text{diag}(q_d)^{-1}.$$

Positive row-and-column scaling preserves the sign of every principal minor. Thus $-J^C$ is a P -matrix if and only if $\tilde{A}$ is.

To verify strict row diagonal dominance, suppress the firm subscript and set $g = 1 - \theta$ and $L = g - \beta s$. Since the off-diagonal multiplier is common across $k \neq d$ and $\sum_{k \neq d} s_k = 1 - s$,

$$\sum_{k \neq d} |\tilde{A}_{dk}| = \varphi |L| (1 - s),$$

and direct simplification gives the row margin $m(s) := \tilde{A}_{dd} - \sum_{k \neq d} |\tilde{A}_{dk}|$ as

$$m(s) = \begin{cases} g \left(\frac{2}{\sigma} - \frac{1}{\rho} + 2\varphi s \right) + 2\varphi \beta s(1 - s), & L \geq 0, \\ \frac{g}{\rho}, & L < 0. \end{cases}$$

For interior shares, $g > 0$. If $L < 0$, the second expression is strictly positive. If $L \geq 0$, the restriction $\sigma \leq 2\rho$ makes $2/\sigma - 1/\rho \geq 0$; the displayed expression is strictly positive for $s \in (0, 1)$, including the boundary case $\sigma = 2\rho$ because then $\varphi > 0$. Hence $\tilde{A}$ is strictly row diagonally dominant with positive diagonal. Every principal submatrix inherits these properties; its Gershgorin discs lie in the open right half-plane, so its determinant is

positive. Thus $\tilde{A}$ —and therefore $-J^C$ —is a P -matrix and J^C is nonsingular. This is the local-regularity fact used in Corollary 1; uniqueness itself was established by the share-space argument. $\square$

Classic related results include Novshek (1985) and Rosen’s concave-game analysis (Rosen, 1965); neither is invoked in the uniqueness proof of Lemma 2.

Corollary 1 uses the uniqueness result in Lemma 2 to obtain comparative statics with respect to unit costs and their structural shifters.

Corollary 1 (Comparative statics). *Under Lemma 2, for each firm d , $\partial q_d^*/\partial C_d < 0$ and $\partial s_d^*/\partial C_d < 0$. Every rival share rises. The focal firm’s markup falls strictly when $\sigma > \rho$ and is unchanged when $\sigma = \rho$; rival quantities have no global sign. With $C_d^D = (\tau_d/z_d)P_d^M$, focal quantity and share fall with τ_d or P_d^M and rise with z_d .*

Proof. Use the share-space representation $C_i = \Lambda f(s_i)$ from Lemma 2. Raise C_d while holding every rival cost fixed. At the old Λ , the implied focal share falls and rival shares are unchanged, so their sum is below one. The new adding-up value must therefore satisfy $\Lambda' > \Lambda$: if it did not, every rival share would weakly fall and the focal share would fall strictly, contradicting adding-up. Consequently each rival share rises strictly.

The focal share nevertheless falls. Suppose instead that $s'_d \geq s_d$. Adding-up then requires some rival k to have $s'_k < s_k$. Since f is strictly decreasing,

$$\frac{f(s'_d)}{f(s'_k)} < \frac{f(s_d)}{f(s_k)},$$

whereas the first-order conditions require the left-hand ratio to equal $C'_d/C_k > C_d/C_k$, the right-hand ratio. This contradiction proves $s'_d < s_d$. Since $\Lambda' > \Lambda$,

$$Q' = \left(\frac{\Lambda}{\Lambda'}\right)^\rho < Q,$$

and $q_d = s_d^{\sigma/(\sigma-1)}Q$ falls because both factors fall. Thus $\partial s_d^*/\partial C_d < 0$ and $\partial q_d^*/\partial C_d < 0$.

The markup is $\mu_d = \epsilon_d / (\epsilon_d - 1)$. This is strictly decreasing in ϵ_d . From Lemma 1, when $\sigma > \rho$, perceived elasticity ϵ_d is strictly decreasing in s_d , so the focal markup falls strictly. When $\sigma = \rho$, $\epsilon_d = \sigma$ is constant and the markup is unchanged. For a rival k , s_k rises but Q falls, so $q_k = s_k^{\sigma/(\sigma-1)} Q$ has no global sign. At the calibrated baseline, whose revenue-share statistic is $r^D \approx (0.90, 0.10)$, rival output expands locally, but that direction is numerical and local, not a theorem. The structural-shifter directions follow from $C_d^D = (\tau_d/z_d)P_d^M$. $\square$

The vertical-stage analysis begins with Lemma 3, which characterizes incremental-surplus shape on a single link under the restricted diagnostic; Lemma 4 then applies that shape to one-link bargaining.

Lemma 3 (Flat-cost fixed-elasticity one-link diagnostic). *Fix a link (d, u) and let $\eta > 1$ be the elasticity in P_d^M . Write $C := C_{du}^U$ and $B(p) := \Delta \Pi_d^D(p)$. For the seller-side statement, impose flat supplier cost ($\psi_u = 0$), hold sales on u 's other links fixed when (d, u) is removed, and use the local isoelastic demand diagnostic $q_{du}^U(p) = \Lambda_{du} p^{-\epsilon_{du}}$ with fixed $\epsilon_{du} := \eta(1 - \bar{\omega}_{du}) > 0$. Here $\bar{\omega}_{du}$ is the link share at the reference point, held fixed only in this diagnostic.*

Define the buyer-participation endpoint

$$\bar{p}^D := \sup \{x > C : B(t) > 0 \text{ for every } t \in (C, x)\} \in (C, \infty].$$

Suppose B has a continuously differentiable extension to $p = C$, with $B(C) > 0$, is twice differentiable wherever positive, and satisfies Assumption 2(a) on $\mathcal{P}_D = (C, \bar{p}^D)$. If $\bar{p}^D < \infty$, suppose $B(p) \downarrow 0$ as $p \uparrow \bar{p}^D$. Then:

- (a) **Buyer side.** *The incremental surplus $B(p)$ is positive and strictly decreasing, and is log-concave in p on $\mathcal{P}_D$.*
- (b) **Seller side.** *If $\epsilon_{du} > 1$, set*

$$p^\dagger := \frac{\epsilon_{du}}{\epsilon_{du} - 1} C.$$

The seller's incremental surplus is positive, strictly increasing, and strictly log-concave on $(C, p^\dagger]$. If $0 < \epsilon_{du} \leq 1$, it is positive, strictly increasing, and strictly log-concave on all of (C, ∞) ; no seller-side interior peak exists.

Proof. Assumption 2(a) directly supplies the buyer-side conditions

$$\frac{dB(p)}{dp} < 0, \quad \frac{d^2 \log B(p)}{dp^2} \leq 0 \quad \text{for all } p \in \mathcal{P}_D. \quad (\text{O6.1})$$

(a) *Buyer side.* Equation (O6.1) is exactly the maintained monotonicity and log-concavity restriction for the one-link buyer incremental surplus, holding all other link prices fixed. Equilibrium profit includes rivals' reoptimization as p changes, so no simple envelope equality is used. Positivity follows from the definition of $\mathcal{P}_D$. This proves the buyer-side statement directly.

(b) *Seller side.* Under the lemma's flat-cost, fixed-local-elasticity restriction, severing (d, u) removes only that link's profit. For a given p , the incremental surplus is

$$\Delta \Pi_u^U(p) = (p - C) q_{du}^U(p).$$

At the reference point, the CES within-bundle elasticity is $-\eta(1 - \bar{\omega}_{du})$. The diagnostic holds that local elasticity fixed at $-\epsilon_{du}$, so

$$\frac{\partial \log q_{du}^U}{\partial \log p} = -\epsilon_{du}.$$

Hence the seller's incremental surplus $\Delta \Pi_u^U(p) = \Lambda_{du}(p - C)p^{-\epsilon_{du}}$ has marginal log-return

$$\frac{d}{dp} \log \Delta \Pi_u^U(p) = \frac{1}{p - C} - \frac{\epsilon_{du}}{p}.$$

If $\epsilon_{du} > 1$, this is positive on $(C, p^\dagger)$ and negative beyond, with a unique seller-side maximizer

at $p^\dagger = \frac{\epsilon_{du}}{\epsilon_{du}-1}C$. For log-concavity,

$$\frac{d^2}{dp^2} \left(\log \Delta \Pi_u^U(p) \right) = -\frac{1}{(p-C)^2} + \frac{\epsilon_{du}}{p^2}.$$

For $\epsilon_{du} > 1$ and $p \leq p^\dagger$, $(p-C)/p \leq 1/\epsilon_{du}$, so the displayed second derivative is strictly negative. If $0 < \epsilon_{du} \leq 1$, it is strictly negative for every $p > C$ because $p-C < p$, while the marginal log-return is strictly positive. This proves the seller-side statements. $\square$

The buyer-side log-concavity is the direct restriction in (O6.1). The buyer-participation endpoint may be finite or infinite; no finite indifference price is assumed. The seller-side $p^\dagger$ and log-concavity result is narrower: it applies only to the flat-cost, fixed-local-elasticity diagnostic stated in Lemma 3.

For the calibrated convex-cost multi-link model, the exact seller surplus used in bargaining is instead

$$\begin{aligned} \mathcal{V}_u(\mathcal{S}; p) &= \sum_{(d', u) \in \mathcal{S}} (p_{d'u}^U - C_{d'u}^U) q_{d'u}^U - \frac{1}{2} c_{0u} \psi_u Q_u(\mathcal{S})^2, \\ \Delta \Pi_u^U(p) &= \mathcal{V}_u(\mathcal{L}_u; p) - \mathcal{V}_u(\mathcal{L}_u \setminus \{(d, u)\}; p), \end{aligned}$$

Here $\mathcal{L}_u$ is supplier u 's active-link set and $Q_u(\mathcal{S}) = \sum_{(d', u) \in \mathcal{S}} q_{d'u}^U$. Both evaluations resolve the affected quantities and supplier total. This difference generally does not reduce to $(p - C_{du}^U) q_{du}^U$, so the convex-cost analysis does not claim the closed-form $p^\dagger$ or global one-link log-concavity. It uses the full objective and the local or uniform Jacobian certificates in Lemma 5.

The closed-form shape restrictions in Lemma 3 guarantee a unique bargained price only for the restricted diagnostic formalized in Lemma 4.

Lemma 4 (Flat-cost fixed-elasticity one-link Nash-in-Nash diagnostic). *Fix rivals' prices*

and let $\gamma \in (0, 1)$. Maintain all conditions of Lemma 3, and define

$$\bar{p} = \begin{cases} \min\{p^\dagger, \bar{p}^D\}, & \epsilon_{du} > 1, \\ \bar{p}^D, & 0 < \epsilon_{du} \leq 1. \end{cases}$$

Under the flat-cost, fixed-local-elasticity seller-demand diagnostic, the one-link Nash-in-Nash problem has a unique interior maximizer $p^* \in (C_{du}^U, \bar{p})$. The endpoint $\bar{p}^D$ is allowed to be infinite.

Proof. Let $p \equiv p_{du}^U$ and write the log-Nash objective

$$h(p) := \gamma \log \Delta \Pi_d^D(p) + (1 - \gamma) \log \Delta \Pi_u^U(p).$$

Write $C = C_{du}^U$ and $B(p) = \Delta \Pi_d^D(p)$. By Lemma 3, on $(C, \bar{p})$ both surpluses are positive, $\log \Delta \Pi_u^U$ is strictly concave, and $\log B$ is concave. Therefore, h is *strictly concave* on $(C, \bar{p})$ because it is a positive-weighted sum of a strictly concave and a concave function.

Compute the derivative:

$$h'(p) = \gamma \frac{B'(p)}{B(p)} + (1 - \gamma) \left(\frac{1}{p - C} - \frac{\epsilon_{du}}{p} \right).$$

As $p \downarrow C$, the seller term tends to $+\infty$, while the buyer term has a finite limit because B extends C^1 to C and $B(C) > 0$. Hence $h'(p) \rightarrow +\infty$.

It remains to obtain a negative derivative before the upper endpoint. If $\epsilon_{du} > 1$ and $p^\dagger < \bar{p}^D$, the seller term is zero at $p^\dagger$ while $B'(p^\dagger)/B(p^\dagger) < 0$, so $h'((p^\dagger)^-) < 0$. If $\bar{p}^D < \infty$ and either $0 < \epsilon_{du} \leq 1$ or $(\epsilon_{du} > 1 \text{ and } \bar{p}^D \leq p^\dagger)$, then $\log B(p) \rightarrow -\infty$ at the finite endpoint. Concavity of $\log B$ implies $(\log B)'(p) \rightarrow -\infty$, whereas the seller term remains finite, so h' is negative near $\bar{p}^D$. The equality $\bar{p}^D = p^\dagger$ is covered by the same finite-endpoint argument.

Finally, suppose $0 < \epsilon_{du} \leq 1$ and $\bar{p}^D = \infty$. For any $p_0 > C$, concavity and strict monotonicity give $(\log B)'(p) \leq (\log B)'(p_0) < 0$ for $p \geq p_0$, while $1/(p - C) - \epsilon_{du}/p \rightarrow 0$.

Thus $h'(p) < 0$ for all sufficiently large p .

Continuity now gives a stationary point $p^* \in (C, \bar{p})$. Strict concavity makes it unique and globally maximizing on that interval. No finite buyer-indifference price is used. $\square$

Lemma 4 solves the restricted flat-cost, fixed-local-elasticity one-link diagnostic: there the log-Nash objective is strictly concave and admits a unique interior optimizer. The convex-cost Nash-in-Nash outcome instead uses full firm-level seller surplus and is the joint root of the stacked first-order conditions $\Phi(\mathbf{p}^U) = 0$, one per link. Links interact through P_d^M , induced Cournot quantities, and supplier total cost. Lemma 5 therefore separates the fixed-metric uniform condition sufficient for global uniqueness from the pointwise condition sufficient for local regularity, without extending the diagnostic's closed-form curvature result to the convex multi-link system.

Lemma 5 records the pointwise Jacobian facts used by the implicit-function argument. Remark 4 then states the separate matrix restriction needed to sign off-diagonal inverse entries.

Lemma 5 (NiN Jacobian: invertibility and diagonal positivity). *Fix the active-link set $\mathcal{L}$. Let $g(\mathbf{p}^U)$ be the log-Nash pseudo-gradient stacked over links $L = (d, u) \in \mathcal{L}$, let $J(\mathbf{p}^U) := \partial g(\mathbf{p}^U) / \partial \mathbf{p}^U$, and let $\mathcal{P}$ denote the participation-feasible box. Under Assumptions 1 and 2:*

- (a) *(Negative diagonal under the relevant certificate.) In the restricted flat-cost diagnostic, $J_{LL} < 0$ wherever Lemmas 3–4 apply. In the full convex-cost system, $J_{LL} < 0$ at every point covered by either the uniform certificate in part (b) or the pointwise certificate in part (c); no sign is asserted over the rest of $\mathcal{P}$ without such a certificate.*
- (b) *(Uniform global certificate.) Suppose $\mathcal{P}$ contains an interior zero and there exists a fixed positive diagonal matrix R such that*

$$S_R(\mathbf{p}^U) := \frac{1}{2} \left(R J(\mathbf{p}^U) + J(\mathbf{p}^U)^\top R \right)$$

is negative definite for every $\mathbf{p}^U \in \mathcal{P}$. Then the zero is unique in $\mathcal{P}$.

(c) (Pointwise local certificate.) On the branch $\mathbf{p}^{U*}(\Theta)$ in Assumption 2(c), $S_R(\mathbf{p}^{U*}(\Theta), \Theta) \prec 0$ for a positive diagonal R , so $J(\mathbf{p}^{U*}(\Theta), \Theta)$ is nonsingular, $-J(\mathbf{p}^{U*}(\Theta), \Theta)$ is a P -matrix, and every diagonal entry of $(-J(\mathbf{p}^{U*}(\Theta), \Theta))^{-1}$ is strictly positive:

$$\left[(-J(\mathbf{p}^{U*}(\Theta), \Theta))^{-1}\right]_{LL} = \frac{\det(-J(\mathbf{p}^{U*}(\Theta), \Theta)_{\hat{L}})}{\det(-J(\mathbf{p}^{U*}(\Theta), \Theta))} > 0,$$

a ratio of positive principal minors, where $-J_{\hat{L}}$ deletes row and column L .

Remark 4 (What diagonal stability does *not* imply: M -matrix structure). *Diagonal stability delivers invertibility and the P -matrix property, but not entrywise inverse-positivity. A blanket statement $-J^{-1} \geq 0$ would require $-J$ to be a nonsingular M -matrix and therefore additionally a Z -matrix with nonpositive off-diagonals. Assumption 2(c) does not impose that sign restriction. Consequently, the comparative-static formula must use the specific entries of $-J^{-1}$; their signs are additional local objects rather than consequences of diagonal stability.*

Proof of Lemma 5. (a) In the restricted flat-cost diagnostic, Lemmas 3–4 give

$$J_{LL} = \frac{\partial^2 h_L}{\partial p_L^2} < 0.$$

At any point in the full system where $S_R \prec 0$, let e_L denote the L th coordinate vector. Then $e_L^\top S_R e_L = R_{LL} J_{LL} < 0$ because $R_{LL} > 0$. This proves the negative diagonal directly under either certificate, without importing the flat-cost formula into the convex-cost model.

(b) This is Rosen’s diagonal-strict-concavity condition with a fixed scaling. Together with the assumed interior zero, it gives uniqueness on $\mathcal{P}$. The primitive bounds establish the condition on the dominant links, but do not verify it uniformly on the low-weight shared link; a calibration-point eigenvalue check is not a substitute for this hypothesis.

(c) Assumption 2(c) gives the pointwise certificate on the selected branch. At $\mathbf{p}^{U*}(\Theta)$, it

implies $R(-J(\mathbf{p}^{U*}(\Theta), \Theta)) + (-J(\mathbf{p}^{U*}(\Theta), \Theta))^\top R \succ 0$ with $R \succ 0$ diagonal, so $-J(\mathbf{p}^{U*}(\Theta), \Theta)$ is diagonally (Lyapunov) stable and therefore positive stable. Diagonal stability is inherited by every principal submatrix, and a real positive-stable matrix has positive determinant. Hence every principal minor of $-J(\mathbf{p}^{U*}(\Theta), \Theta)$ is positive, so it is a P -matrix (Berman and Plemmons, 1994); nonsingularity and Cramer's rule give the displayed positive diagonal entries. $\square$

Proof of Proposition 1 (NiN system: uniqueness and comparative statics).

Step 1 (Curvature of the Cournot block). By Lemmas 1 and 2, the Cournot stage has a unique equilibrium for any fixed vector of unit costs. The share-space argument establishes uniqueness without the restriction $\sigma \leq 2\rho$; under that restriction, the elasticity-scaled marginal-profit Jacobian $\tilde{A}$ is strictly row diagonally dominant and nonsingular.

Step 2 (Own-link curvature). Lemmas 3–4 give strict own-link concavity for the restricted flat-cost, fixed-local-elasticity diagnostic. In the full convex-cost system, the relevant negative own derivatives follow at the selected equilibrium from the pointwise certificate, or throughout the feasible box if the uniform certificate is imposed; the proof does not claim global convex-cost concavity from the diagnostic.

Step 3 (Joint system: uniform versus pointwise regularity). Let $\mathcal{L}$ be the set of active links and write $\mathbf{p}^U = (p_L)_{L \in \mathcal{L}}$ with $L = (d, u)$. Define for each link

$$h_L(p_L; \mathbf{p}_{-L}^U) = \gamma_d \log \Delta \Pi_d^D(p_L; \mathbf{p}_{-L}^U) + (1 - \gamma_d) \log \Delta \Pi_u^U(p_L; \mathbf{p}_{-L}^U),$$

the pseudo-gradient $g = (g_L)_L$ with $g_L := \partial h_L / \partial p_L$, and its Jacobian $J := \partial g / \partial \mathbf{p}^U$. Work on the compact box $\mathcal{P} = \prod_L [\underline{p}_L, \bar{p}_L]$ given by Assumption 1.

At points covered by the relevant certificate, own derivatives are negative; cross-effects still need not be uniformly dominated on the low-weight shared (Micron) row. The primitive curvature bounds therefore do not by themselves imply fixed-metric negative definiteness throughout $\mathcal{P}$.

If the uniform condition in Lemma 5(b) is imposed and $\mathcal{P}$ contains an interior zero, Rosen’s diagonal-strict-concavity argument gives a unique zero in $\mathcal{P}$. This is the conditional global result, applicable to flat or convex cost only when that uniform condition is independently established.

For convex cost, Assumption 2(c) imposes $S_R(\mathbf{p}^{U*}) \prec 0$ at the selected zero. Lemma 5(c) then gives a nonsingular Jacobian, the P -matrix property, and local uniqueness of the branch by the implicit-function theorem. This pointwise condition does not prove global uniqueness.

O7 Proof of Proposition 2

This proof fixes the active-link set and separates analytic regularity from signs that require calibration. Lemma 5(c) supplies the pointwise Jacobian certificate used by the implicit-function argument, and any signed claims remain local to the calibrated branch.

Step 1 (Regularity and derivative formula). Fix the active set of links and work on the participation interior of the compact box $\mathcal{P}$ from Assumption 1. Lemmas 1–2 make the Cournot equilibrium map $C^D \mapsto q^D(C^D)$ continuously differentiable. The CES index is smooth in positive link prices, and the full supplier profit in equation (10) combines those smooth demand maps with polynomial total cost. Because both with-link and without-link incremental surpluses are strictly positive on this interior, their logs are continuously differentiable (and twice differentiable wherever the Jacobian is evaluated). Hence each link objective h_L and the pseudo-gradient

$$g(\mathbf{p}^U, \Theta) = \left(g_L(\mathbf{p}^U, \Theta) \right)_{L \in \mathcal{L}} = \left(\frac{\partial h_L}{\partial p_L}(\mathbf{p}^U, \Theta) \right)_{L \in \mathcal{L}}$$

are C^1 in $(\mathbf{p}^U, \Theta)$ for any collection of primitives Θ (e.g. $\gamma, \tau, \mathbf{z}, \dots$).

On the continuation branch in Assumption 2(c), Proposition 1 and Lemma 5 imply that the Jacobian

$$J(\mathbf{p}^U, \Theta) = \frac{\partial g(\mathbf{p}^U, \Theta)}{\partial \mathbf{p}^U}$$

is invertible at $\mathbf{p}^{U*}$ ($-J$ being a P -matrix). Therefore, by the Implicit Function Theorem, there exist neighborhoods $\mathcal{V}$ of Θ and $\mathcal{U}$ of $\mathbf{p}^{U*}$ and a C^1 map $\phi : \mathcal{V} \rightarrow \mathcal{U}$ such that

$$g(\phi(\Theta), \Theta) = 0, \quad \phi(\Theta_0) = \mathbf{p}^{U*}(\Theta_0).$$

Differentiating $g(\phi(\Theta), \Theta) \equiv 0$ with respect to any scalar primitive Θ_k yields

$$\frac{\partial \mathbf{p}^{U*}}{\partial \Theta_k} = -J(\mathbf{p}^{U*}(\Theta), \Theta)^{-1} \frac{\partial g(\mathbf{p}^U, \Theta)}{\partial \Theta_k} \Big|_{\mathbf{p}^U = \mathbf{p}^{U*}(\Theta)}.$$

Assumption 2(c) and Lemma 5(c) further show that $-J(\mathbf{p}^{U*}(\Theta), \Theta)^{-1}$ has a strictly positive diagonal on this local branch. Signed elasticities therefore use the specific entries of $-J^{-1}$, subject to Remark 4.

Step 2 (Buyer-specific wedges and productivity). Fix a buyer d and an active link $L = (d, u)$ with $\omega_{du} \in (\varepsilon, 1 - \varepsilon)$ and $p_{du}^{U*} - C_{du}^U > 0$ at the baseline. On the box $\mathcal{P}$ and in a neighbourhood of Θ , the local derivative formulas in Step 1 apply with $\Theta_k \in \{\ln \tau_d, \ln z_d\}$.

By construction, the unit cost for the buyer d satisfies $C_d^D \propto \tau_d/z_d$. Holding the active set fixed, log changes in τ_d or z_d therefore operate through the same cost channel, with opposite signs. Differentiating the one-link first-order condition $g_L(\mathbf{p}^U, \Theta) = 0$ with respect to $\ln \tau_d$ or $\ln z_d$, keeping rival link prices temporarily fixed, yields expressions of the form

$$\frac{\partial g_L}{\partial \ln \tau_d} = \Xi_{d,u}^{(\tau)}(\Theta), \quad \frac{\partial g_L}{\partial \ln z_d} = \Xi_{d,u}^{(z)}(\Theta),$$

where $\Xi_{d,u}^{(\tau)}$ and $\Xi_{d,u}^{(z)}$ are continuous functions of primitives and equilibrium objects (shares, markups, demand elasticities). No additional global sign restrictions on these objects follow from the curvature inequalities alone: buyer surplus and seller surplus both respond to C_d^D and p_{du}^U , and the net effect depends on their relative magnitudes at the calibrated equilibrium.

Stacking across all links and applying the derivative formula in Step 1, the elasticities for

the focal link can be written as

$$\begin{aligned}\frac{\partial \ln p_{du}^{U*}}{\partial \ln \tau_d} &= \Psi_{d,u}^{(\tau)}(\Theta) := \frac{1}{p_{du}^{U*}} \left[-J(\mathbf{p}^{U*}, \Theta)^{-1} \frac{\partial g(\mathbf{p}^{U*}, \Theta)}{\partial \ln \tau_d} \right]_L, \\ \frac{\partial \ln p_{du}^{U*}}{\partial \ln z_d} &= \Psi_{d,u}^{(z)}(\Theta) := \frac{1}{p_{du}^{U*}} \left[-J(\mathbf{p}^{U*}, \Theta)^{-1} \frac{\partial g(\mathbf{p}^{U*}, \Theta)}{\partial \ln z_d} \right]_L,\end{aligned}$$

where $[\cdot]_L$ denotes the L th component. Because J^{-1} is continuous wherever J is invertible (Lemma 5) and the shock vectors $\partial g / \partial \ln \tau_d$ and $\partial g / \partial \ln z_d$ are continuous in $(\mathbf{p}^{U*}, \Theta)$, the functions $\Psi_{d,u}^{(\tau)}$ and $\Psi_{d,u}^{(z)}$ are continuous in Θ on the region where the active set is unchanged. The same representation applies to any rival link $L' = (d', u')$ with $d' \neq d$, yielding well-defined elasticities $\partial \ln p_{d'u'}^{U*} / \partial \ln \tau_d$ and $\partial \ln p_{d'u'}^{U*} / \partial \ln z_d$.

The proposition therefore does not impose a global sign on $\Psi_{d,u}^{(\tau)}$ or $\Psi_{d,u}^{(z)}$ from primitives alone: their signs are determined locally at the calibrated equilibrium through the shock vectors and the specific entries of $-J^{-1}$ (not sign-definite, Remark 4). This completes the proof of Proposition 2.

Additional bargaining-weight comparative statics

Consider now a change in the bargaining weight γ_d for buyer d . For any link $L = (d, u)$ attached to d ,

$$g_L(\mathbf{p}^U, \Theta) = \gamma_d \frac{\partial \log \Delta \Pi_d^D}{\partial p_{du}^U} + (1 - \gamma_d) \frac{\partial \log \Delta \Pi_u^U}{\partial p_{du}^U},$$

where $\Delta \Pi_d^D$ and $\Delta \Pi_u^U$ are buyer and seller incremental surpluses on link L . Differentiating with respect to γ_d gives

$$\frac{\partial g_L}{\partial \gamma_d} = \frac{\partial}{\partial p_{du}^U} (\log \Delta \Pi_d^D - \log \Delta \Pi_u^U).$$

In the restricted flat-cost diagnostic, Lemma 3 gives a decreasing buyer surplus and an increasing seller surplus. In the convex-cost system, these derivatives instead depend on the

full with-link and without-link profits. If, at a selected point,

$$\frac{\partial}{\partial p_{du}^U} \log \Delta \Pi_d^D < 0, \quad \frac{\partial}{\partial p_{du}^U} \log \Delta \Pi_u^U > 0,$$

then, at that point,

$$\frac{\partial g_L}{\partial \gamma_d} = \frac{\partial}{\partial p_{du}^U} \log \Delta \Pi_d^D - \frac{\partial}{\partial p_{du}^U} \log \Delta \Pi_u^U < 0.$$

For any link $L' = (d', u')$ with $d' \neq d$, the FOC $g_{L'}$ does not depend directly on γ_d , so $\partial g_{L'}/\partial \gamma_d = 0$.

Stacking across links and using the derivative formula from Step 1,

$$\frac{\partial \mathbf{p}^{U*}}{\partial \gamma_d} = -J(\mathbf{p}^{U*}, \Theta)^{-1} \frac{\partial g(\mathbf{p}^{U*}, \Theta)}{\partial \gamma_d}.$$

Under this conditional surplus-derivative sign, the shock vector $\partial g/\partial \gamma_d$ has negative entries on links attached to d and zeros on links attached only to other buyers. Even then, the positive diagonal of $-J^{-1}$ in Lemma 5(c) does not by itself sign a focal-link response because within-buyer off-diagonal terms also enter. Rival-link responses likewise propagate through off-diagonal inverse-Jacobian entries whose signs are unrestricted by Remark 4. Thus both own- and rival-link bargaining-weight responses remain local and network-conditional unless their full current-calibration derivatives are separately evaluated.

Flat-versus-convex response accounting

The full convex-cost surplus changes both the pseudo-gradient Jacobian and the wedge-shock vector. Because with-link and without-link quantities re-solve and the Nash objective uses the log of incremental surplus, the resulting change is not a per-supplier rank-one update and does not admit a primitive “bargaining versus convex cost” Woodbury decomposition. Remark 5 records the exact accounting identity that remains.

Remark 5 (Exact accounting identity, not a channel decomposition). *Let (J_0, b_0) denote the Jacobian and wedge-shock vector on the selected flat-cost branch and (J_ζ, b_ζ) their counterparts on the selected convex-cost branch, where $b_r = \partial g_r / \partial \ln \tau_d$. Define $\Delta J = J_\zeta - J_0$ and $\Delta b = b_\zeta - b_0$. Whenever both Jacobians are nonsingular, the two implicit-function responses satisfy the exact identity*

$$-J_\zeta^{-1} b_\zeta = -J_0^{-1} b_0 - J_0^{-1} \Delta b + J_0^{-1} \Delta J J_\zeta^{-1} b_\zeta.$$

The last two terms jointly collect changes in full with-link and disagreement profits, equilibrium quantities, log-surplus denominators, and the shock vector. They are accounting residuals, not separately identified economic channels, and no low-rank claim is made. The quantitative marginal-cost-versus-Nash-in-Nash comparison uses separate full equilibrium solves and does not rely on this identity.

Remark 6 (Network-routed corrections). *The maintained conditions impose no entrywise sign restriction on J_0^{-1} , J_ζ^{-1} , ΔJ , or Δb . Flat-versus-convex differences therefore carry no network-free sign. The index-level spillover in Corollary 1 remains exact CES accounting conditional on the solved link responses; it is not derived from graph connectedness or a low-rank convex-cost update.*

O8 Buyer-side curvature and numerical evidence

This section separates the analytic buyer-curvature result in Lemma 6 from the calibrated nested-CES evidence summarized in Remark 8 and Table O5.3. The former is confined to the linear-Cournot full-incremental benchmark; the CES signs are numerical, network-conditional results from full-equilibrium solutions.

Lemma 6 (Buyer-side log-submodularity: linear Cournot). *Suppose $D = 2$ and downstream Cournot competition takes place under linear inverse demand $p^D = a - b \sum_{d'} q_{d'}^D$, $a, b > 0$.*

If $\Pi_{d,-u}^D = 0$ (buyer d exits when link (d, u) is removed), then for all parameters consistent with positive equilibrium profits ($a > 2c_d - c_{d'}$),

$$\frac{\partial^2 \log \Delta \Pi_d^D}{\partial (\ln C_d^D)^2} < 0.$$

This establishes negative buyer-side log-curvature, and hence log-submodularity by (15), in the full-incremental special case.

Proof of Lemma 6. With unit costs $c_d = (\tau_d/z_d)p_L$ and $c_{d'}$ fixed, the two-firm linear Cournot equilibrium gives $\Pi_d^D = (a - 2c_d + c_{d'})^2/(9b)$. Since $\Pi_{d,-u}^D = 0$, $\Delta \Pi_d^D = \Pi_d^D$. Setting $\varpi \equiv a - 2c_d + c_{d'} > 0$:

$$\log \Delta \Pi_d^D = \text{const} + 2 \log \varpi.$$

Differentiating twice with respect to $\ln c_d$ (using $\partial \varpi / \partial \ln c_d = -2c_d$):

$$\frac{\partial \log \Delta \Pi_d^D}{\partial \ln c_d} = \frac{-4c_d}{\varpi}, \quad \frac{\partial^2 \log \Delta \Pi_d^D}{\partial (\ln c_d)^2} = \frac{-4c_d}{\varpi} \left(1 + \frac{2c_d}{\varpi} \right) < 0.$$

The inequality holds since $c_d, \varpi > 0$. Under the full-incremental restriction of (15), the cross-derivative has the same sign because its CES chain factor is positive; in this one-input case it is $1/p_L$. The buyer-side claim follows. $\square$

Numerical illustration. To make the mechanism in Lemma 6 concrete, consider the linear-Cournot benchmark with parameters $a = 10$, $b = 1$, and unit costs $c_{\text{NV}} = 2$, $c_{\text{AMD}} = 3$. The pre-shock Cournot margin for NVIDIA is $\varpi \equiv a - 2c_{\text{NV}} + c_{\text{AMD}} = 9$ and profit $\Pi_{\text{NV}}^D = \varpi^2/9 = 9$. The log-derivative of profit with respect to log-cost is $\partial \ln \Pi_{\text{NV}}^D / \partial \ln c_{\text{NV}} = -4c_{\text{NV}}/\varpi = -8/9 \approx -0.889$. Now suppose the export wedge rises so that c_{NV} increases by 1% (holding p_L fixed momentarily). The Cournot margin compresses to $\varpi' = 8.96$ and profit falls to $(8.96)^2/9 \approx 8.920$. The log-derivative becomes $-4 \cdot 2.02/8.96 \approx -0.902$: the buyer's marginal sensitivity to input costs has *increased in magnitude*. This is the log-submodularity condition in action—the wedge amplifies how much NVIDIA cares about

each additional dollar of HBM cost. The second-order curvature $\partial^2 \ln \Pi^D / \partial (\ln c)^2 \approx -1.28$ (at baseline) is unambiguously negative. This is a buyer-side result; the negotiated-price response additionally depends on the numerical Nash-in-Nash log-Nash routine and network.

Remark 7 (No general nested-CES curvature theorem). *Let $x = \ln C$ and use the equilibrium identity $\Pi = Cq/(\epsilon - 1)$. Then*

$$\frac{d^2 \log \Pi}{dx^2} = \frac{d^2 \log q}{dx^2} - \frac{\epsilon_{xx}}{\epsilon - 1} + \frac{\epsilon_x^2}{(\epsilon - 1)^2}.$$

The condition $\epsilon_x > 0$ alone does not sign this expression because the curvature of equilibrium quantity and the second derivative of perceived elasticity also enter. Thus no general nested-CES negative-curvature result follows from the maintained primitives. Equilibrium profit includes rivals' reoptimization, so the single-agent envelope equality does not apply.

Remark 8 (CES case). *Only Lemma 6, the linear-Cournot full-incremental benchmark, supplies an analytic buyer-curvature result. Remark 7 explains why no universal nested-CES curvature theorem is available. In the calibrated multi-supplier model, the reported negotiated-price direction is instead numerical and based on full-equilibrium solutions. Table O5.3 reports the complete elasticity grid, recalibration protocol, and reporting screen in Section O5.*

In the calibrated CES model, the numerical Nash-in-Nash log-Nash routine and network yield a calibration-specific negotiated-price response. Under Scenario A ($\tau_{NV} + 10\%$, $\tau_{AMD} + 5\%$) at the baseline $(\sigma, \rho, \eta) = (6, 4, 2.5)$, SK hynix and Micron together yield $100 \Delta \ln P_{NV}^M \approx -3.6\%$ versus $100 \Delta \ln \mu_{NV} \approx -0.12\%$. Section 5 defines the scale and mixed-unit interpretation of the reported cushioning magnitude; Table O5.3 contains the full grid comparisons.

O9 Constrained-Cournot equilibrium under an output cap

At each point of the quantity-instrument counterfactual in Section 5, the two Scenario A wedges are removed and both restricted buyers receive ceilings equal to their own equilibrium outputs on the matched 2:1 cost-wedge path. Thus NVIDIA’s headline ceiling is about 21% below its baseline output, while AMD’s ceiling is its corresponding Scenario A output. For each capped buyer d , the problem $\max_{q_d \leq \bar{q}_d} \Pi_d^D$ gives the complementarity

$$q_d = \min\{q_d^{\text{FOC}}, \bar{q}_d\}, \quad \lambda_d = \max\{0, MR_d(\bar{q}_d) - C_d^D\} \geq 0,$$

where q_d^{FOC} is the unconstrained best response and λ_d the shadow value of the cap; uncapped firms best-respond to the capped quantities. The upstream Nash-in-Nash problem is unchanged in form—each active link maximizes the log-Nash product of the two incremental surpluses, with disagreement payoffs evaluated at the constrained equilibrium, which also respects the cap. The absorber weakens *endogenously*: with q_d pinned, severing a link changes the buyer’s profit only through the fixed-scale bundle cost (no volume-expansion term), and the shared supplier’s total-volume incentive is muted, leaving only within-bundle substitution. Along the reported calibrated path, the full-NiN input-price index rises as the cap tightens; this numerical result is consistent with weaker volume preservation and a positive upstream margin. Complementarity alone does not imply $\lambda_d \rightarrow 0$ or global monotonicity, and no theorem is claimed.

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